

**Kalinga Institute of Industrial Technology Deemed to be
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MSC/ADAMS Lab**

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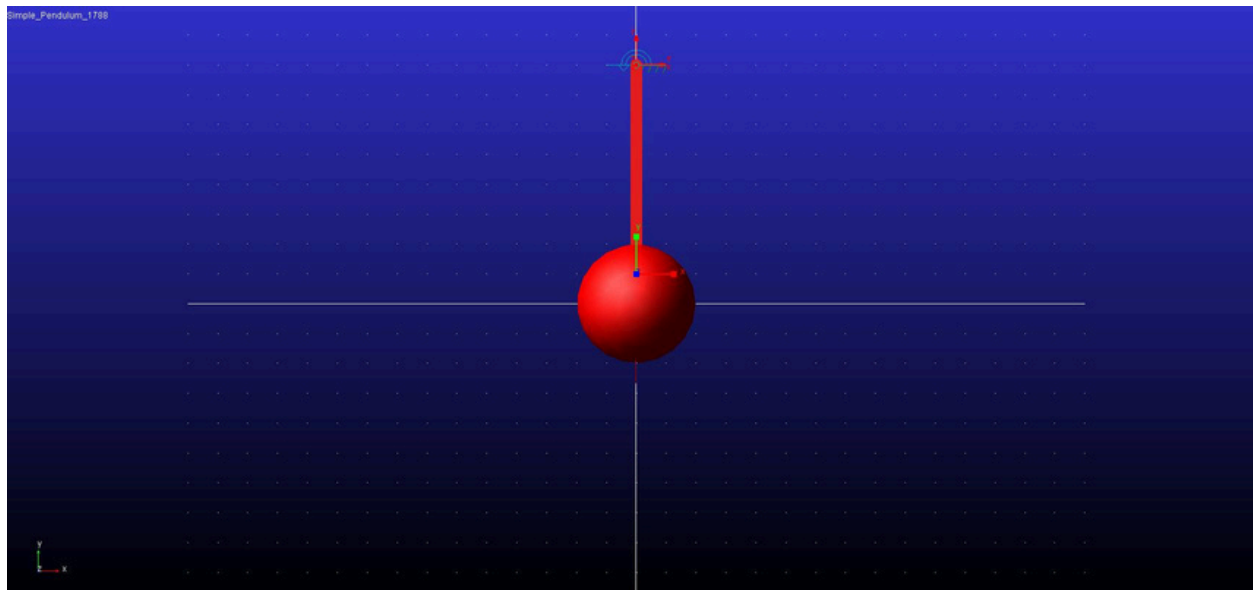
TOPIC:SIMPLE PENDULUM

Section 1: Mechanism Description

Section 1.1 Picture of the mechanism

A **simple pendulum** consists of:

- A point mass (bob)
- A light, inextensible string or rigid rod
- A fixed pivot point
- Gravity acting downward



Section 1.2 Purpose of the mechanism

The purpose of a simple pendulum is to:

- Demonstrate oscillatory motion
- Study time-period and harmonic motion
- Serve as a basic model for vibrations systems

Applications:

- Timekeeping devices (clocks)
- Seismometers
- Physics and dynamics studies

Section 1.3 Kinematic sketch

The kinematic sketch shows:

- A mass suspended from a fixed pivot
- Motion constrained along a circular arc
- Angular displacement measured from vertical

Degrees of Freedom (DOF): **1 (angular)**

Section 1.4 Expected operation

- When displaced and released, the pendulum oscillates about its equilibrium position.
- Motion occurs due to gravitational restoring force.
- For small angles, motion approximates **simple harmonic motion**.

Operating speed:

- Low-speed oscillatory motion

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
------------------	------------------

String length (L)	40cm
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Bob diameter	20cm
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Section 2.1.2 Masses of every component

Component	Mass
------------------	-------------

Bob	1kg
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String	Negligible
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Pivot	Negligible
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Section 2.1.3 Moments of inertia

Component	Moment of Inertia
------------------	--------------------------

Bob (point mass)	$I = mL^2$
------------------	------------

String	Neglected
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Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- Pivotfixedto ground
- Stringlengthconstant
- Motionconfinedtoaverticalplane
- Nolateralorout-of-planemotion

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- Gravityactingonthemass

Section 2.3.2 Output component

- Angulardisplacementofpendulum
- Angularvelocityandacceleration

Section 3: Kinematic simulation

Equation of motion

$$\ddot{\theta} + \frac{g}{L} \sin\theta = 0$$

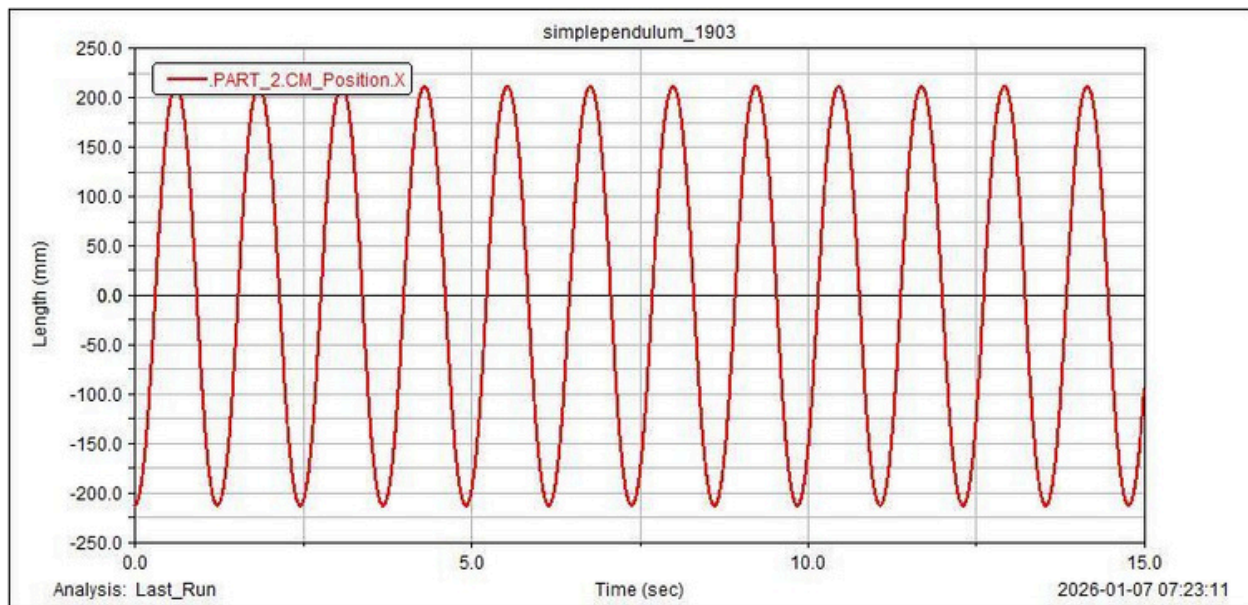
For small angles:

$$\ddot{\theta} + \frac{g}{L} \theta = 0$$

Section 3.1 Output Component

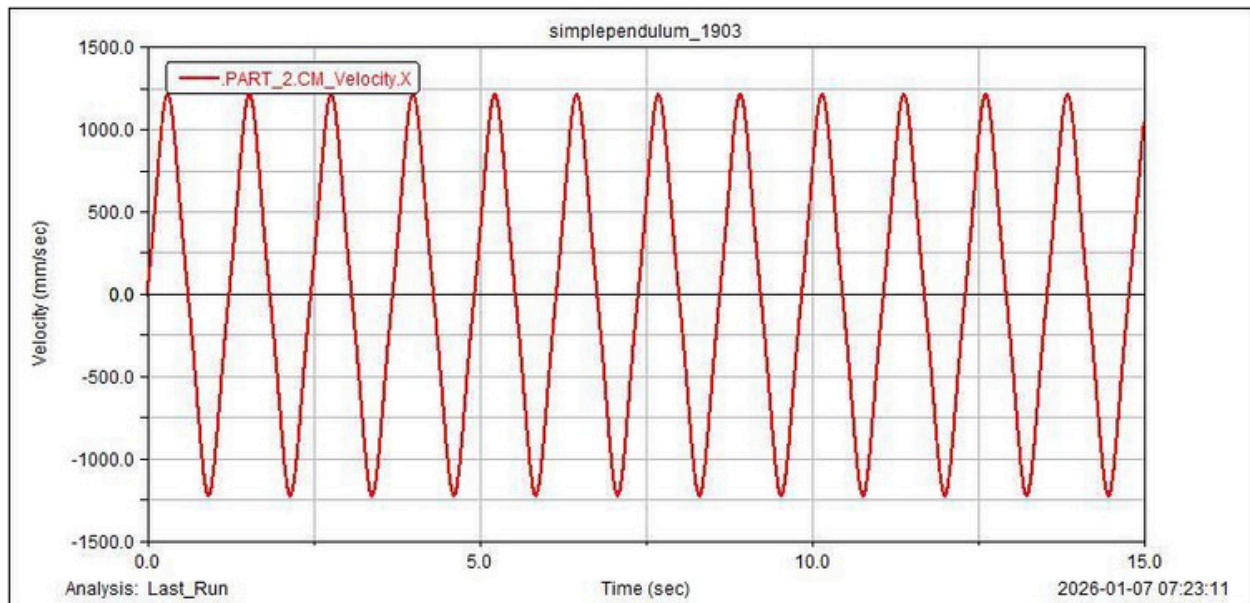
Section 3.1.1 Angular displacement profile

- Plot: $\theta(t)$ vs Time
- Sinusoidal for small angles



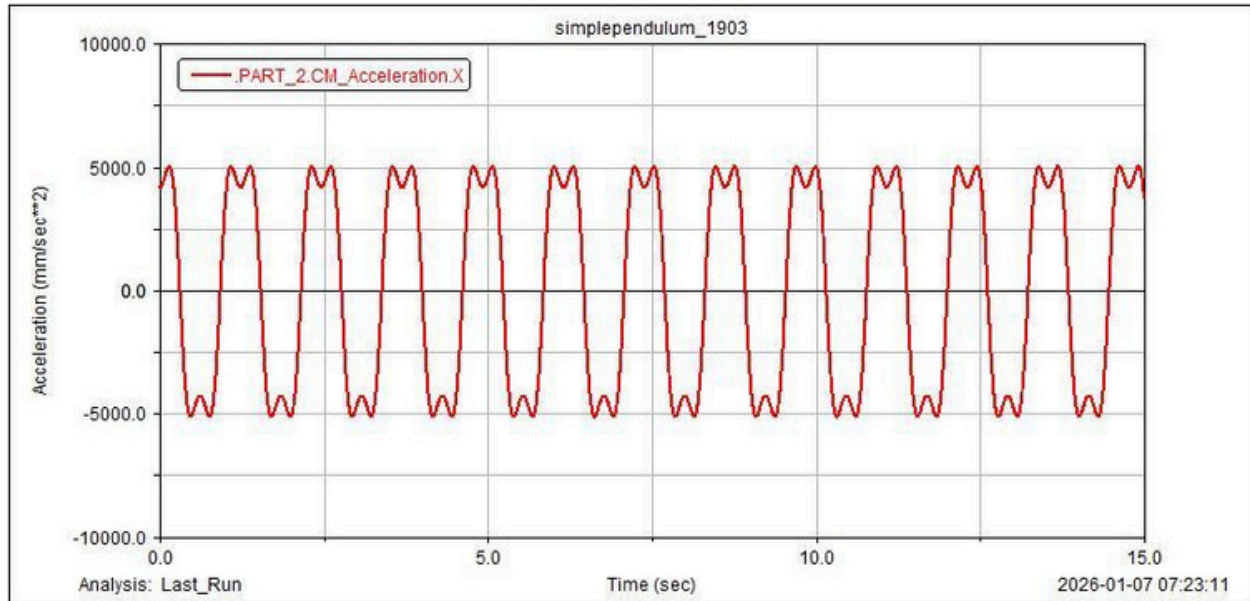
Section 3.1.2 Angular velocity profile

- Plot: $\dot{\theta}(t)$ vs Time
- Maximum at mean position



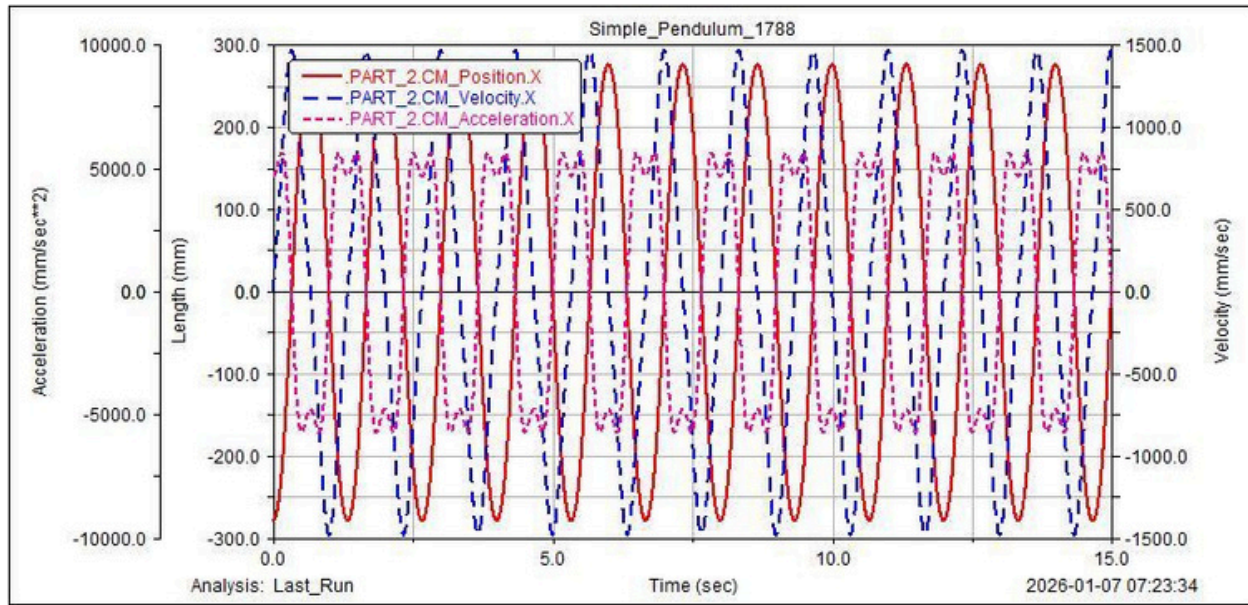
Section 3.1.3 Angular acceleration profile

- Plot: $\ddot{\theta}(t)$ vs Time
- Maximum/minimum positions



Section 3.1.4 Plot for all

- Angular displacement
- Angular velocity
- Angular acceleration



Section 3.1.5 Comments

- Time period independent of mass
- Increasing length increases time period
- Small-angle approximation gives accurate results

Section 4: Dynamic Motion Analysis

Section 4.1 Initial Angular Displacement

Section 4.1.1 Small initial angle

Let:

$$\theta_0 = 10^\circ$$

Plot:

- Angular displacement vs Time
- Angular velocity vs Time
- Angular acceleration vs Time

Section 4.1.2 Double initial angle

$$\theta_0 = 20^\circ$$

- Superimpose plots
- Observe deviation from perfect sinusoidal motion

Section 4.2 External Periodic Torque

Section 4.2.1 Sinusoidal torque

$$T(t) = T_0 \sin(\omega t)$$

Plot:

- Angular velocity vs Time
- Angular acceleration vs Time

Section 4.2.2 Double sinusoidal torque

$$T(t) = 2T_0 \sin(\omega t)$$

- Superimpose results
- Observe increase in oscillation amplitude

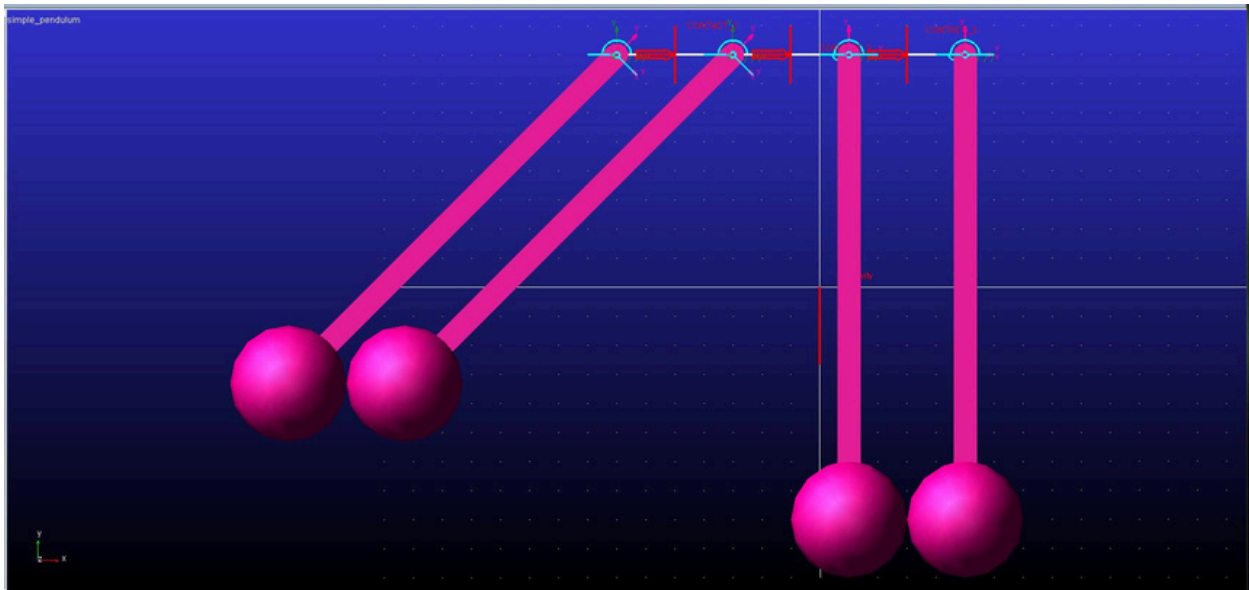
TOPIC: NEWTON'S 3RD LAW

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

The mechanism consists of:

- **Four identical pendulum bobs**
- **Equal-length strings**
- **A rigid support frame**
- **Bobs aligned in a straight line and touching each other**



Section 1.2 Purpose of the mechanism

The purpose of this mechanism is to:

- **Demonstrate Newton's Third Law of Motion**
- **Show action–reaction force pairs**
- **Illustrate momentum and energy transfer during collisions**

Applications:

- Physics education
- Impact force studies
- Conservation law demonstrations

Section 1.3 Kinematic sketch

The kinematic sketch shows:

- Four pendulums suspended from a common frame
- Motion constrained to a vertical plane
- Collisions occurring along a single horizontal line

Degrees of Freedom (DOF):

- **1 DOF per pendulum**
- Total system DOF = 4

Section 1.4 Expected operation

- When one pendulum is displaced and released, it strikes the adjacent pendulum.
- The impact force is transmitted through the stationary pendulums.
- The last pendulum swings out with nearly equal velocity.
- This demonstrates **equal and opposite forces** during interaction.

Operating speed:

- Low-speed oscillatory and impact motion

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
String length (L)	100 cm
Bob diameter	20cm
Spacing between bobs	Zero (touching)

Section 2.1.2 Masses of every component

Component	Mass
Each pendulum bob	0.5 kg
Strings	Negligible
Frame	Fixed

Section 2.1.3 Moments of inertia

Component	Moment of Inertia
Each bob (point mass)	$I = mL^2$
Strings	Neglected

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- Strings are inextensible
- Motion restricted to a single vertical plane
- Frame is fixed
- Bobs collide elastically along one axis

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- Initial angular displacement of one pendulum
- Gravitational force

Section 2.3.2 Output component

- Angular displacement of remaining pendulums
- Impact forces between bobs

Section 3: Kinematic simulation

Equation of motion (individual pendulum)

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0$$

Collision condition

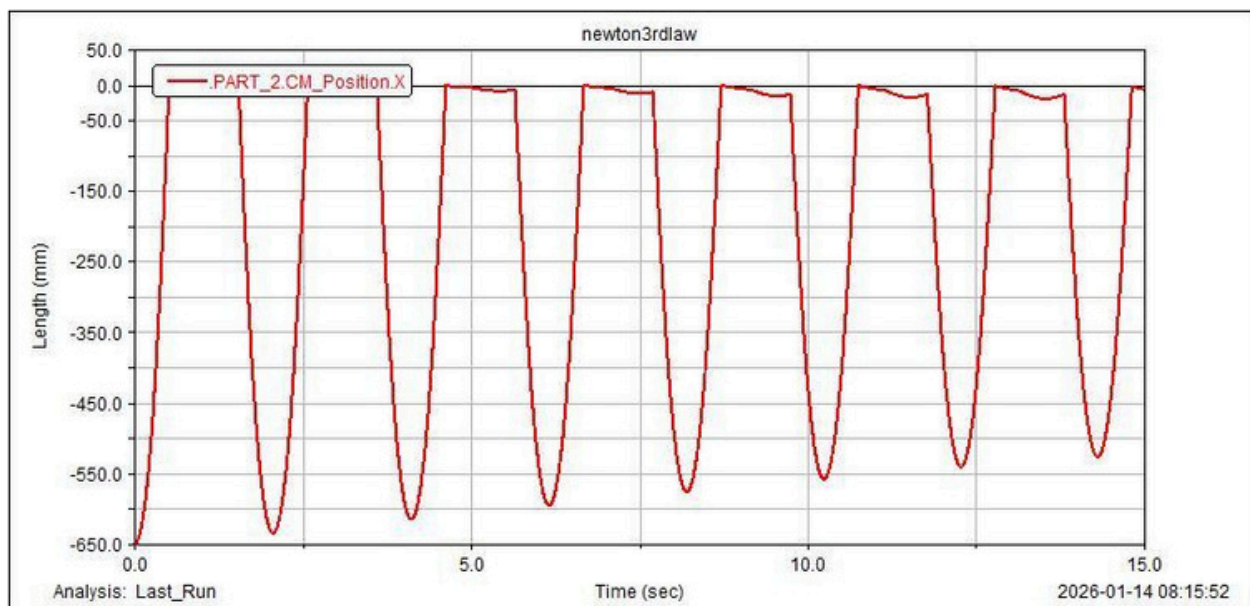
- Linear momentum conserved
- Contact forces satisfy:

$$F_{12} = -F_{21}$$

Section 3.1 Output Component

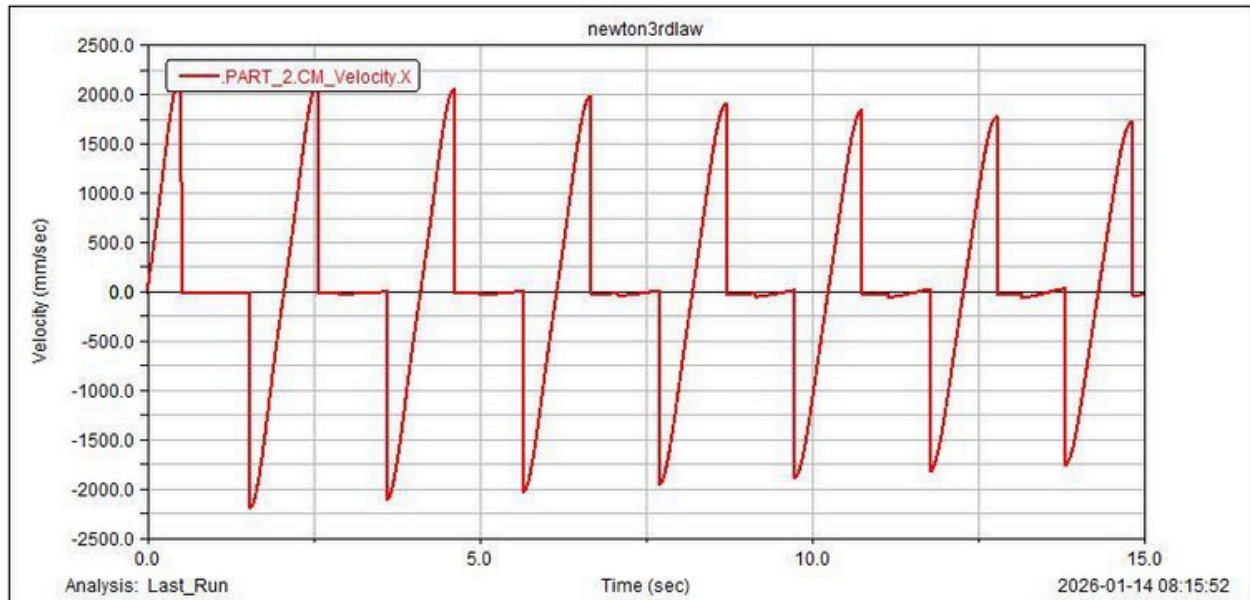
Section 3.1.1 Angular displacement profile

- Plot: Angular displacement vs Time for all four pendulums
- Show transfer of motion from first to last pendulum



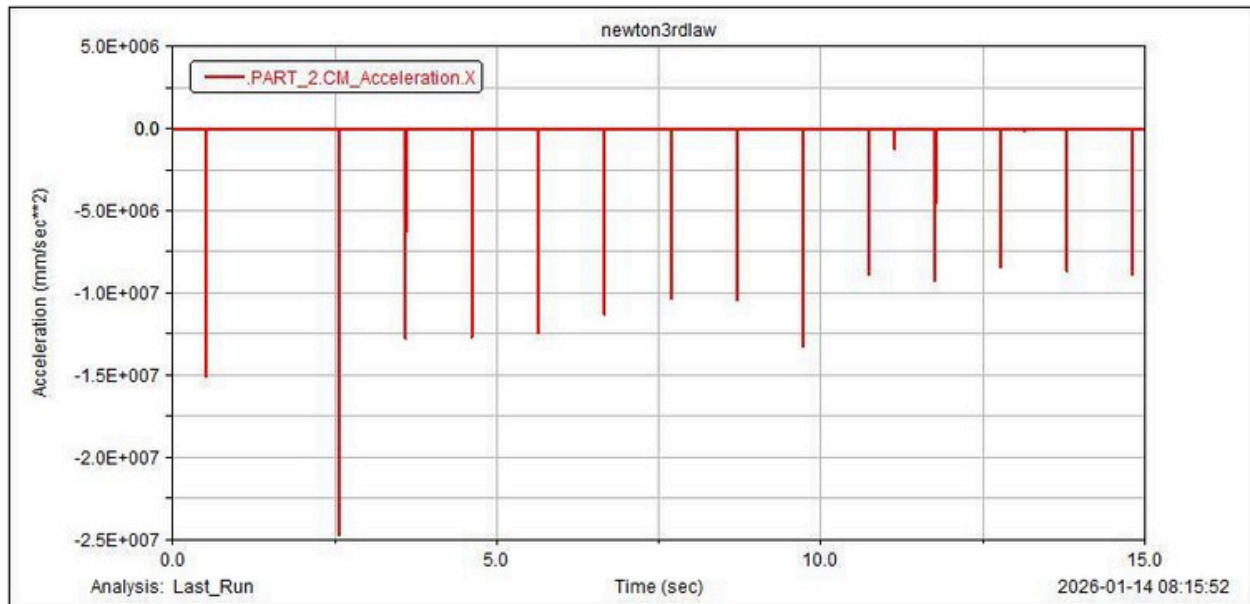
Section 3.1.2 Angular velocity profile

- Plot: Angular velocity vs Time
- Equal magnitude velocities before and after collision



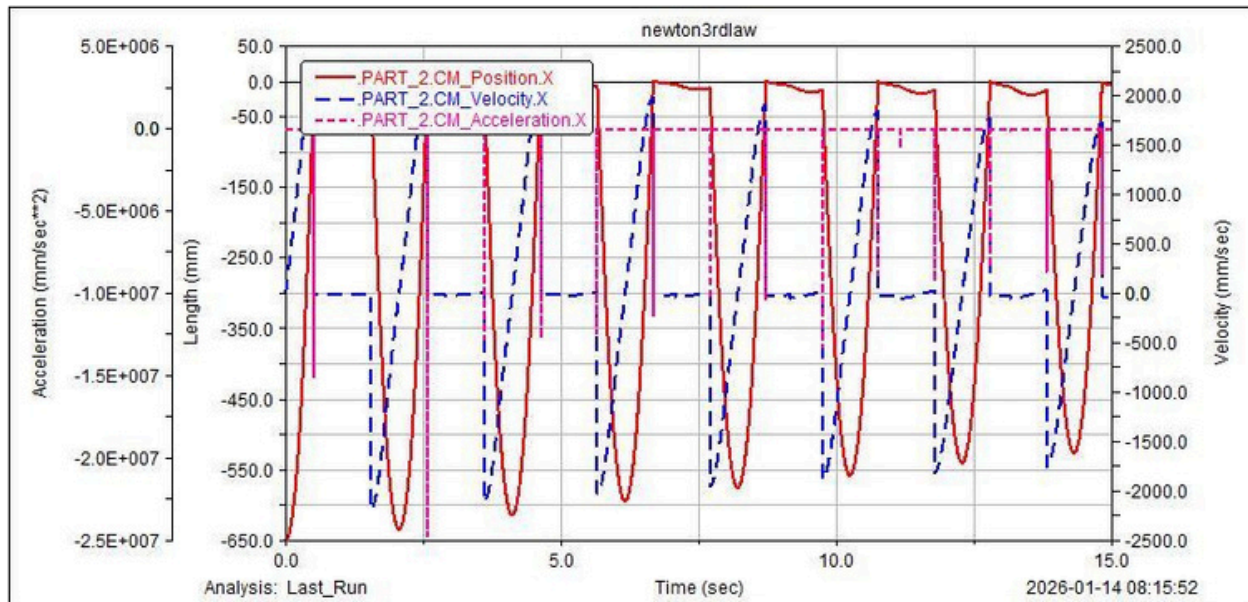
Section 3.1.3 Angular acceleration profile

- Plot: Angular acceleration vs Time
- Sharp acceleration peaks during impact



Section 3.1.4 Plot for all

- Angular displacement
- Angular velocity
- Angular acceleration



Section 3.1.5 Comments

- Middle pendulums remain nearly stationary
- Motion transfer demonstrates Newton's Third Law
- Small energy losses due to damping and air resistance

Section 4: Dynamic Motion Analysis

Section 4.1 Single Pendulum Release

Section 4.1.1 One pendulum displaced

Let:

$$\theta_0 = 15^\circ$$

Plot:

- Angular displacement vs Time
- Contact force vs Time
- Angular velocity vs Time

Section 4.1.2 Double pendulum release

Two pendulums released simultaneously:

$$\theta_0 = 15^\circ$$

- Two pendulums swing out on opposite sides
- Superimpose results with single pendulum case

Section 4.2 Energy and Force Analysis

Section 4.2.1 Force interaction

- Measure force during collision
- Show:

$$F_{\text{action}} = -F_{\text{reaction}}$$

Section 4.2.2 Energy transfer

- Plot kinetic energy vs Time
- Observe near-conservation of energy

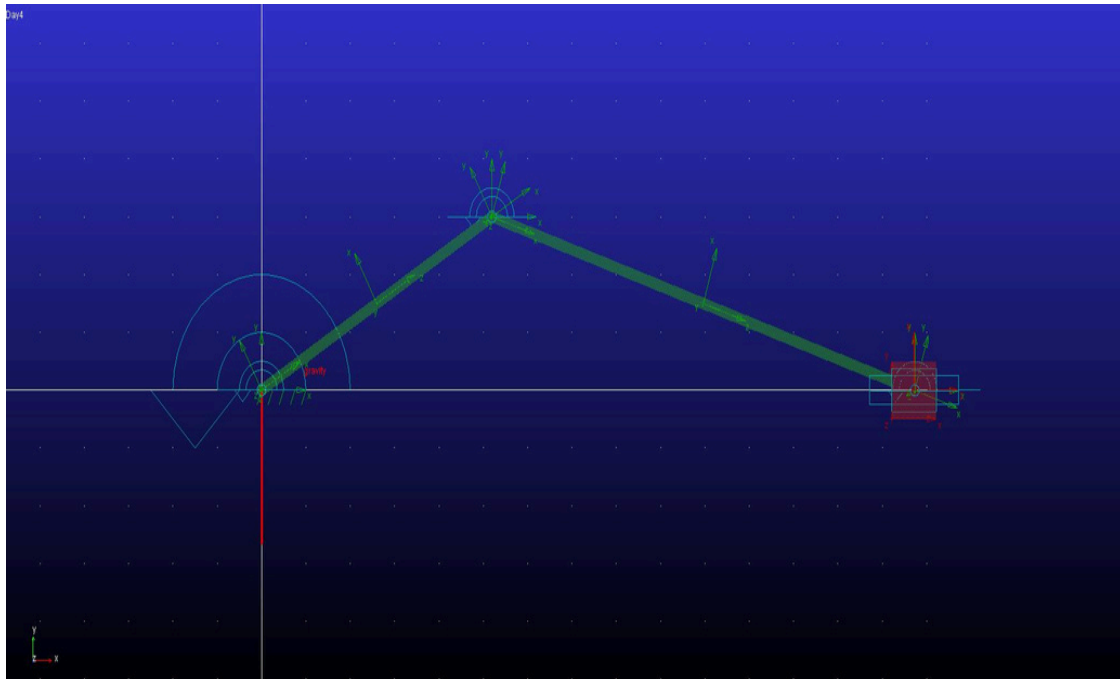
TOPIC: SLIDER-CRANK MECHANISM

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

A slider-crank mechanism consists of:

- Crank
- Connectingrod
- Slider(piston)
- Fixedframe(ground)



Section 1.2 Purpose of the mechanism

The purpose of a slider-crank mechanism is to:

- Convert **rotary motion into linear motion** or vice versa

- Transmit power between rotating and reciprocating components

Common applications:

- Internal combustion engines
- Reciprocating pumps
- Compressors
- Shaping machines

Section 1.3 Kinematic sketch

The kinematic sketch shows:

- A rotating crank pinned to the frame
- A connecting rod linking crank to slider
- A slider constrained to move along a straight line

Degrees of Freedom (DOF): 1

Section 1.4 Expected operation

- Rotation of the crank causes the slider to move back and forth.
- The slider reaches maximum displacement at **top dead center (TDC)** and **bottom dead center (BDC)**.
- Motion is periodic and depends on crank speed and link lengths.

Operating speed:

- Medium to high speed (engine-type motion)

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
------------------	------------------

Crank length (r)	30cm
------------------	------

Connecting rod length (l)	50cm
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Slider width	10cm
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Section 2.1.2 Masses of every component

Component	Mass
------------------	-------------

Crank	0.8 kg
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Connecting rod	1.2 kg
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Slider	2.0 kg
--------	--------

Section 2.1.3 Moments of inertia

Component	Moment of Inertia
------------------	--------------------------

Crank	$I = \frac{1}{2}mr^2$
-------	-----------------------

Connecting rod	$I = \frac{1}{12}ml^2$
----------------	------------------------

Slider	Neglected
--------	-----------

(pure translation)

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- Crank: rotational motion about fixed pin
- Connecting rod: planar motion
- Slider: constrained to linear motion only
- Frame: fixed reference

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- Crank driven by:
 - Constant angular velocity
 - Applied torque

Section 2.3.2 Output component

- Slider displacement, velocity, and acceleration

Section 3: Kinematic simulation

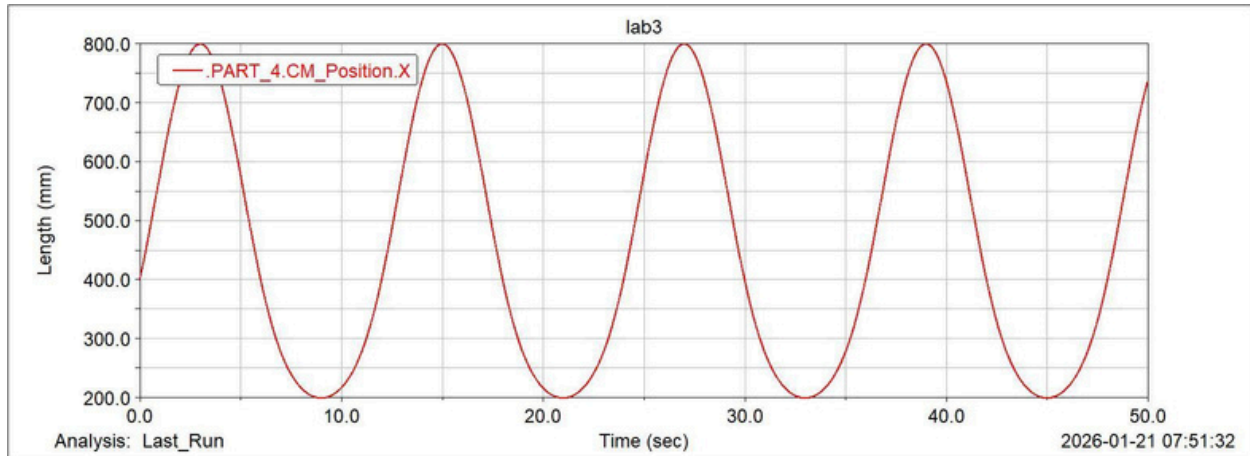
Slider displacement equation

$$x = r \cos \theta + l_2 - r_2 \sin 2\theta$$

Section 3.1 Output Component

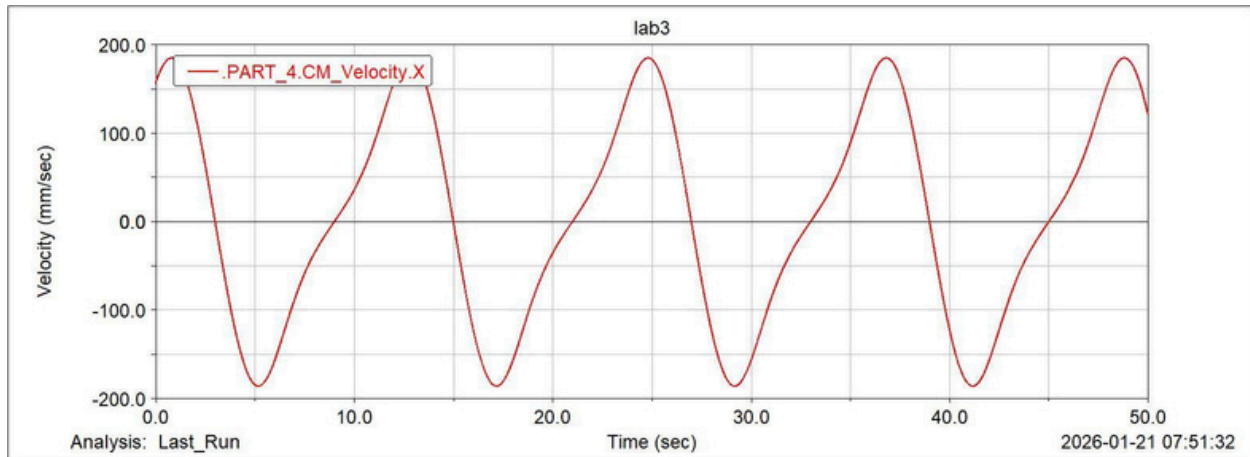
Section 3.1.1 Displacement profile

- Plot: Slider displacement vs Crank angle/Time
- Motion is non-sinusoidal



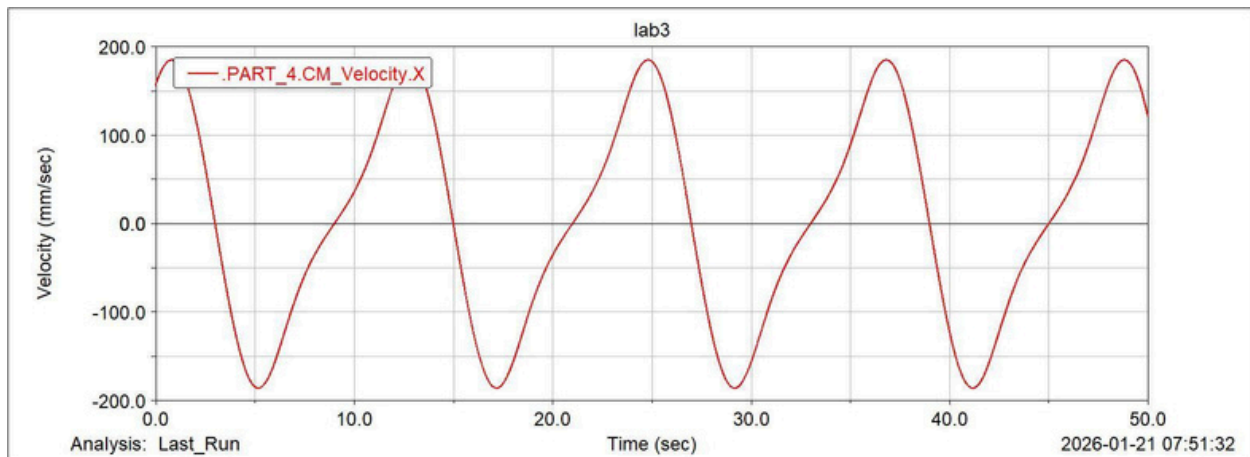
Section 3.1.2 Velocity profile

- Plot:SlidervelocityvsTime
- Maximumvelocityoccursbeforemid-stroke



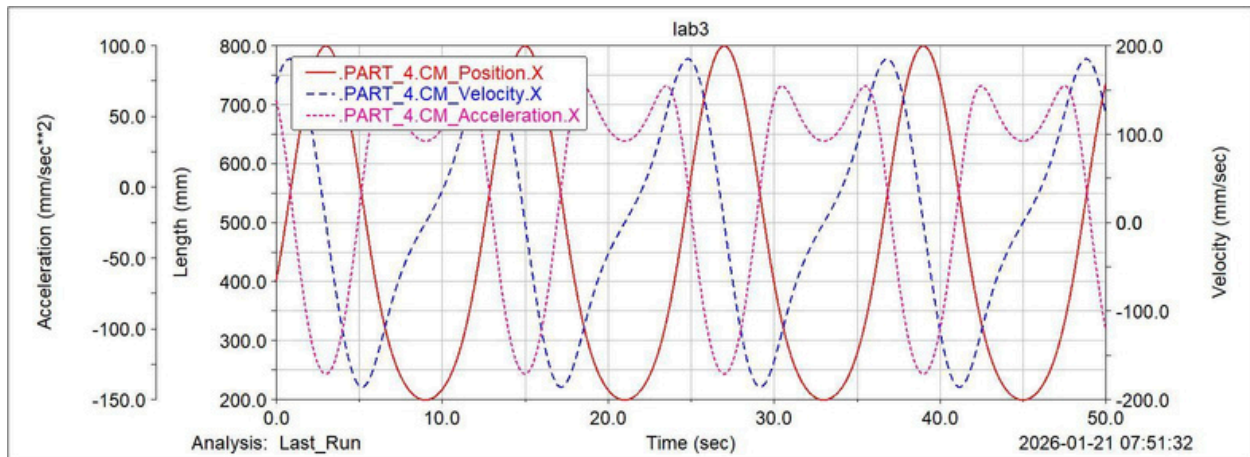
Section 3.1.3 Acceleration profile

- Plot:SlideraccelerationvsTime
- Largeaccelerationneardeadcenters



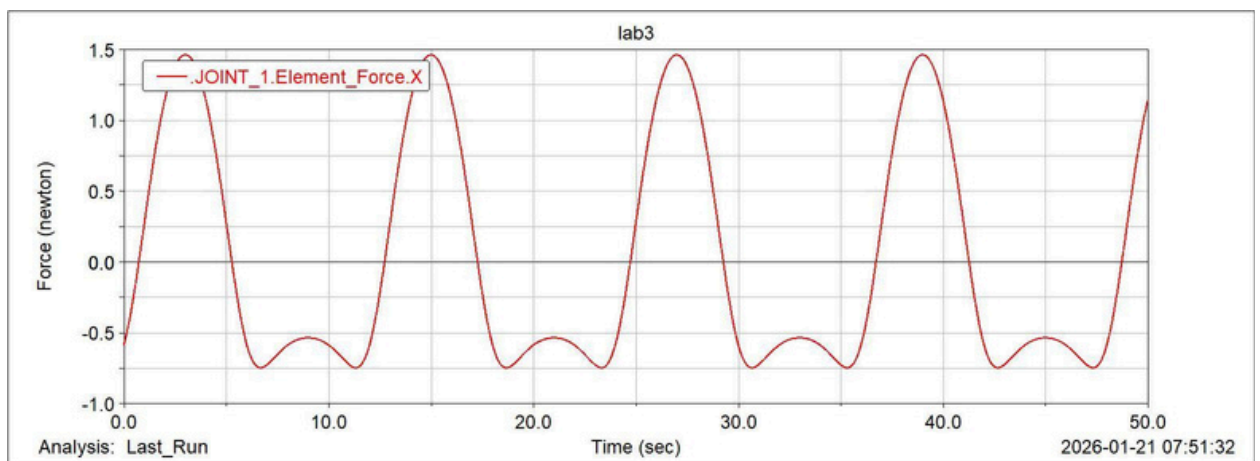
Section 3.1.4 Plot for all

- Angular displacement
- Angular velocity
- Angular acceleration

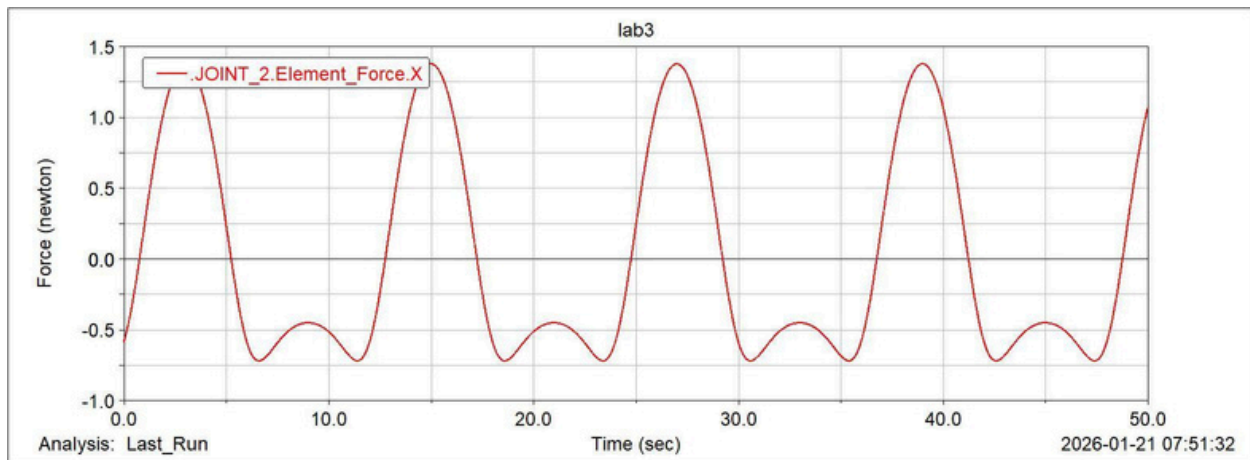


Section 3.1.5 Force

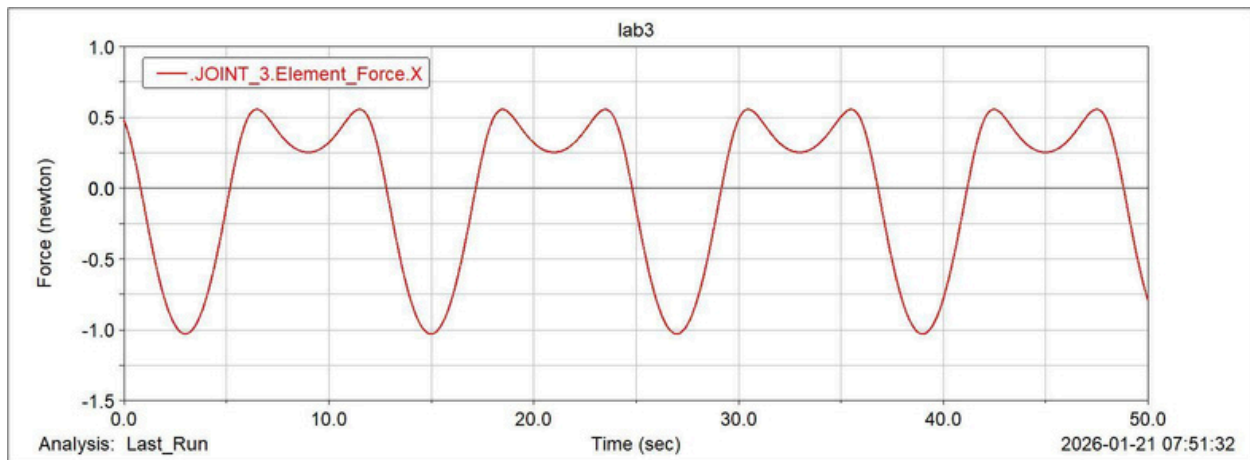
- Joint 1



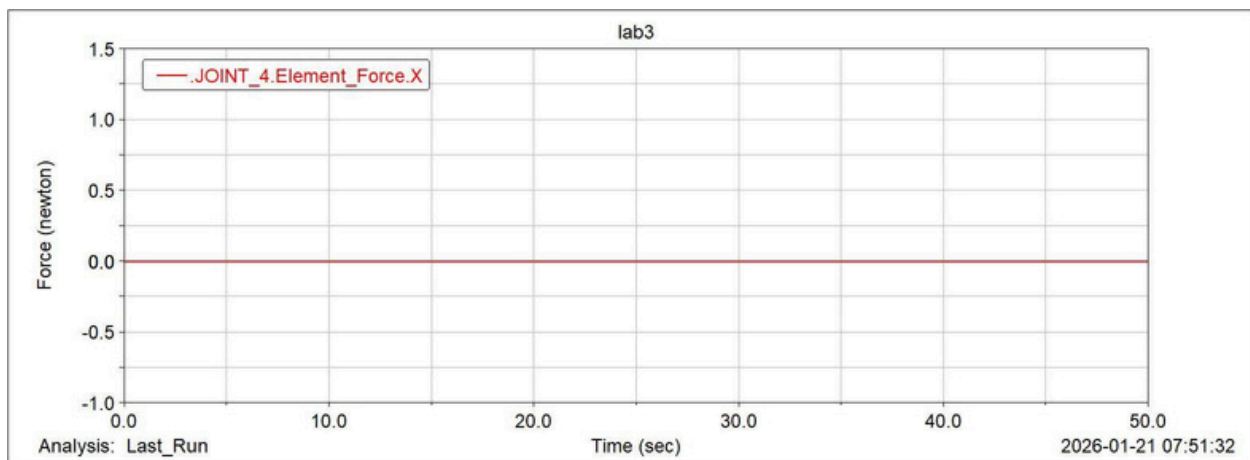
- Joint 2



- Joint 3



- Joint 4



Section 3.1.6 Comments

- Increasing crank speed increases velocity and acceleration
- Longer connecting rod reduces acceleration peaks
- Motion is asymmetric about mid-stroke

Section 4: Dynamic Motion Analysis

Section 4.1 Constant Driving Torque

Section 4.1.1 Constant torque

Let:

$T = 10 \text{ Nm}$

Plot:

- Slider velocity vs Time
- Slider acceleration vs Time
- Reaction forces at joints

Section 4.1.2 Double constant torque

$T = 20 \text{ Nm}$

- Superimpose plots
- Observe increase in slider speed and inertia forces

Section 4.2 Sinusoidal Driving Torque

Section 4.2.1 Sinusoidal torque

$$T(t) = T_0 \sin(\omega t)$$

Plot:

- Slider velocity vs Time
- Slider acceleration vs Time

Section 4.2.2 Double sinusoidal torque

$$T(t) = 2T_0 \sin(\omega t)$$

- Superimpose plots
- Observe force amplification at high speeds

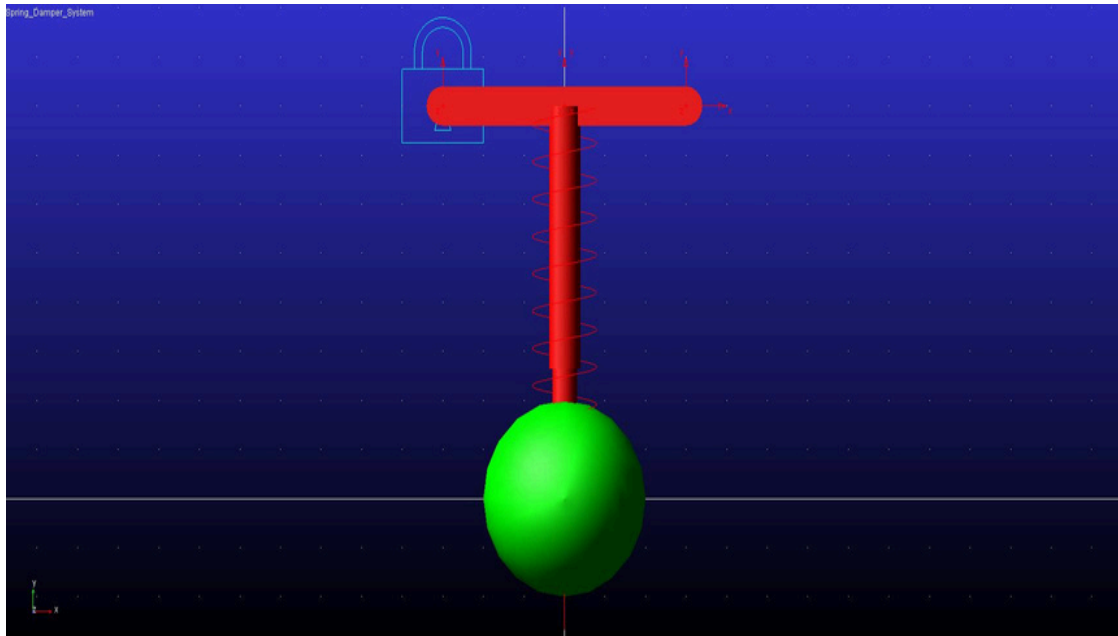
TOPIC:SPRING DAMPER SYSTEM

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

A single degree of freedom (SDOF) spring-damper system consisting of:

- A rigid mass m
- A linear spring with stiffness k
- A viscous damper with damping coefficient c
- Fixed rigid support (ground)



Section 1.2 Purpose of the mechanism

The purpose of a spring–damper system is to:

- Absorb vibrations
- Reduce oscillations
- Control motion of mechanical systems

Such systems are widely used in:

- Vehicle suspension systems
- Vibration isolation
- Shock absorbers
- Structural vibration control

Section 1.3 Kinematic sketch

The kinematic sketch consists of:

- One translational mass moving along a single axis
- Spring and damper connected in parallel between the mass and ground
- Motion constrained to one direction only

Degrees of Freedom (DOF): 1 (translational)

Section 1.4 Expected operation

- When an external force is applied, the mass oscillates.
- The spring provides a restoring force proportional to displacement.
- The damper dissipates energy proportional to velocity.
- The system eventually settles to equilibrium depending on damping.

Operating speed:

Low-speed mechanical vibration (typically < 10 Hz in simulation).

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
Sphere Radius	10cm
Spring free length	80cm
Damper length	80cm
Rod diameter	80cm

Section 2.1.2 Masses of every component

Component	Mass
------------------	-------------

Sphere	2kg
--------	-----

Spring	Negligible
--------	------------

Damper	0.5 kg
--------	--------

Section 2.1.3 Moments of inertia

Since motion is purely translational:

- Rotational inertia is **neglected**
- Only lumped mass is considered

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- Mass constrained to move only along the x-axis
- Spring and damper fixed to ground at one end
- No rotation allowed

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- External force $F(t)$ applied to the mass

Section 2.3.2 Output component

- Displacement of the mass $x(t)$

Section 3: Kinematic simulation

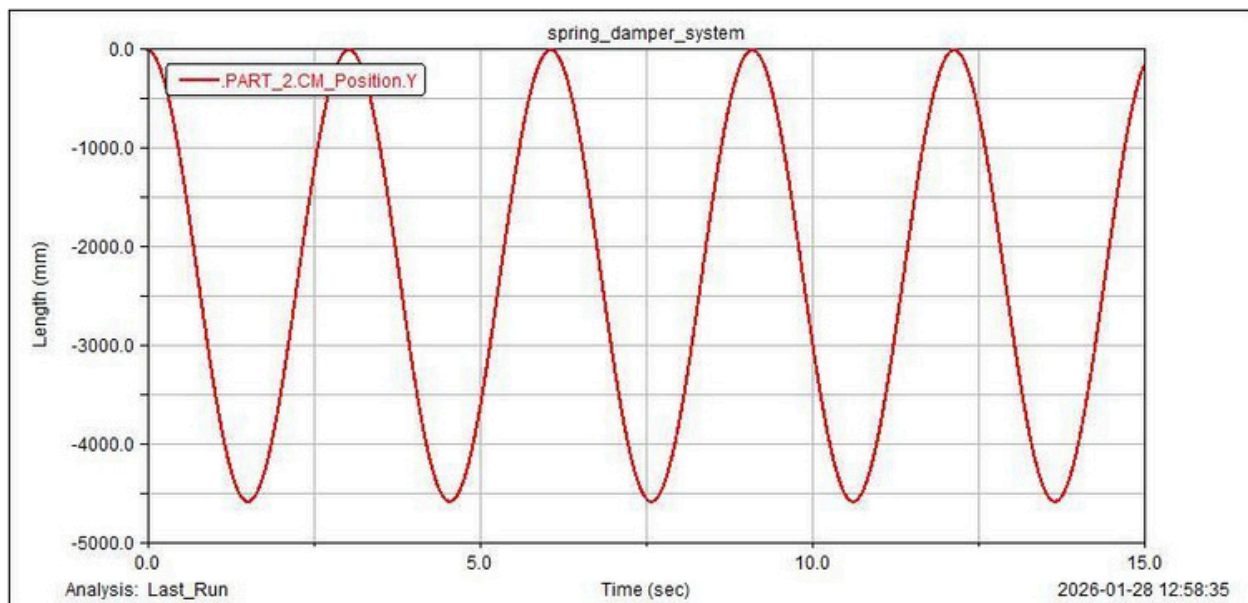
Governing equation of motion:

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

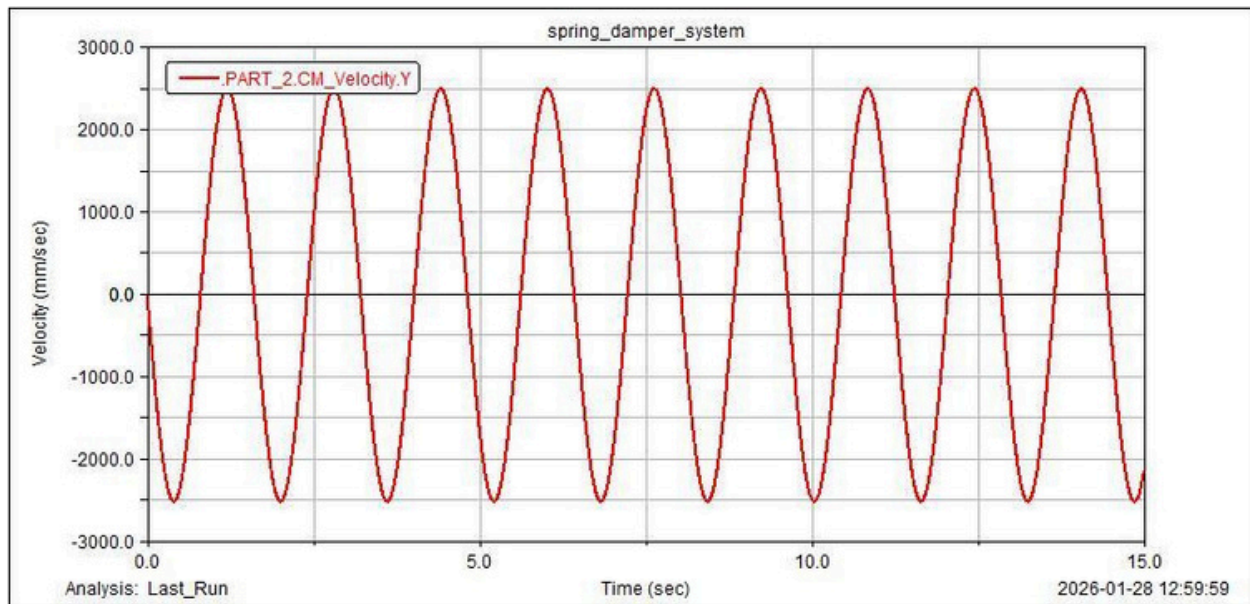
Section 3.1 Output Component

Section 3.1.1 Displacement profile

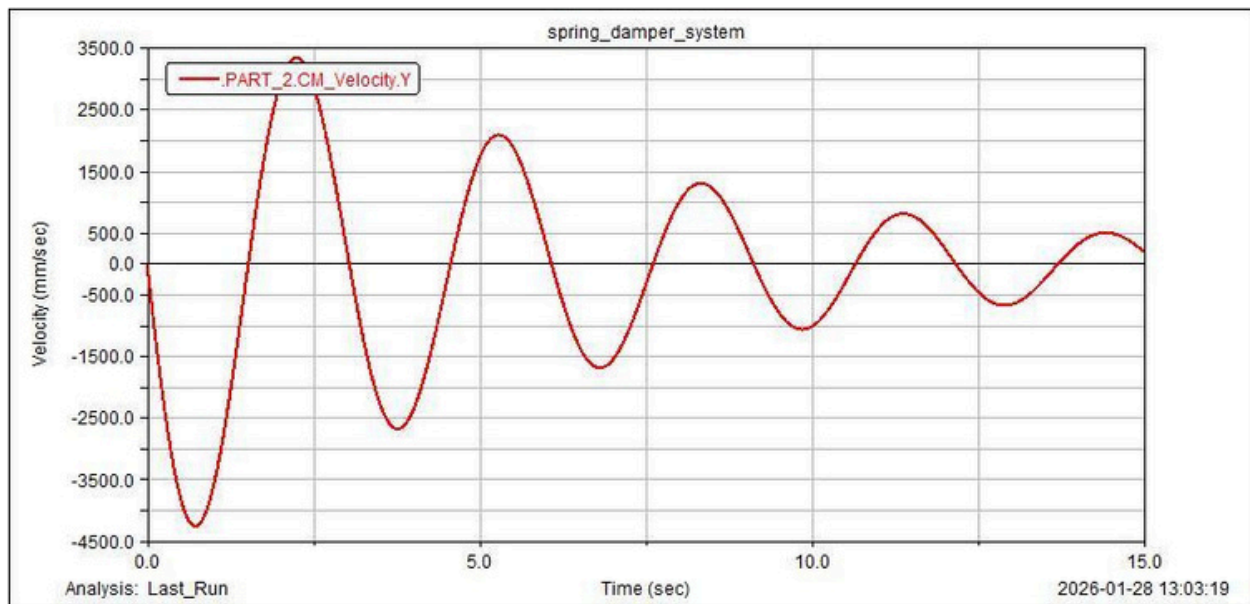
- Plot: $x(t)$ vs Time
- Shows oscillatory decay based on damping
- Without Damping Coefficient More Stiffness Coefficient



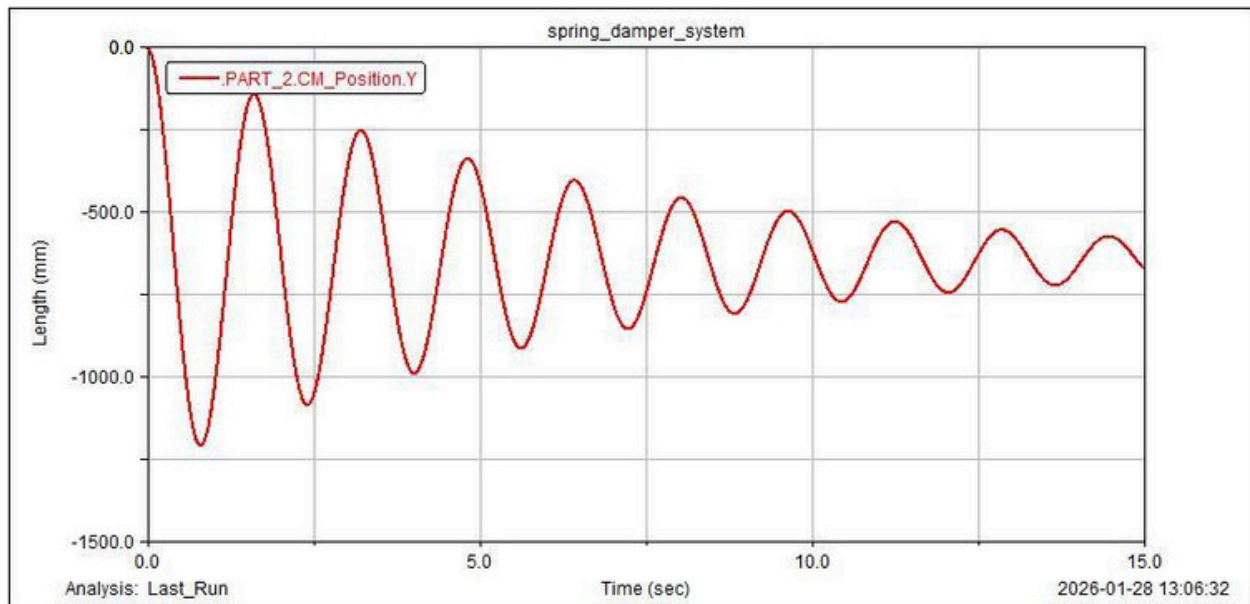
- WithoutDampingCoefficientLessStiffnessCoefficient



- WithDampingCoefficientMoreStiffnessCoefficient

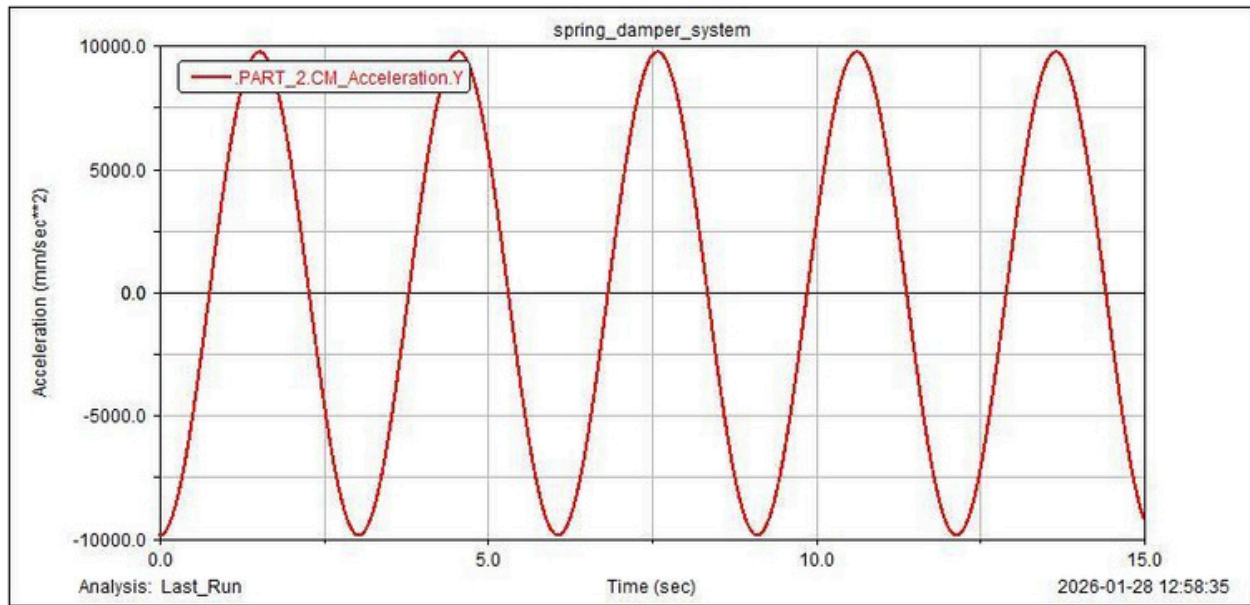


- WithDampingCoefficientLessStiffnessCoefficient

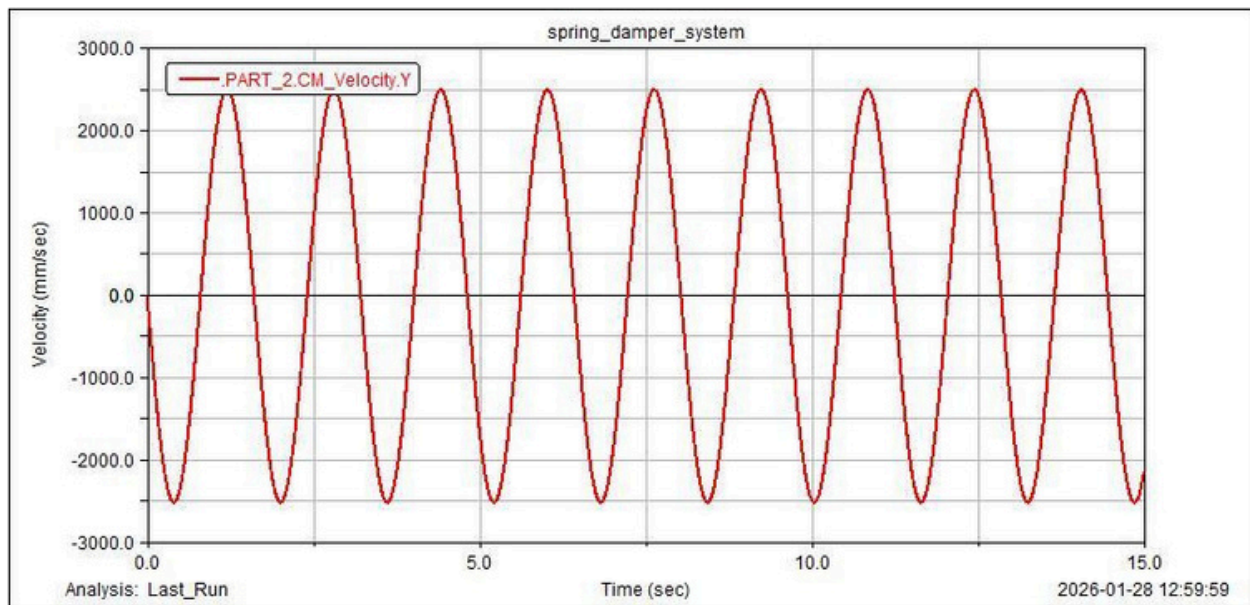


Section 3.1.2 Velocity profile

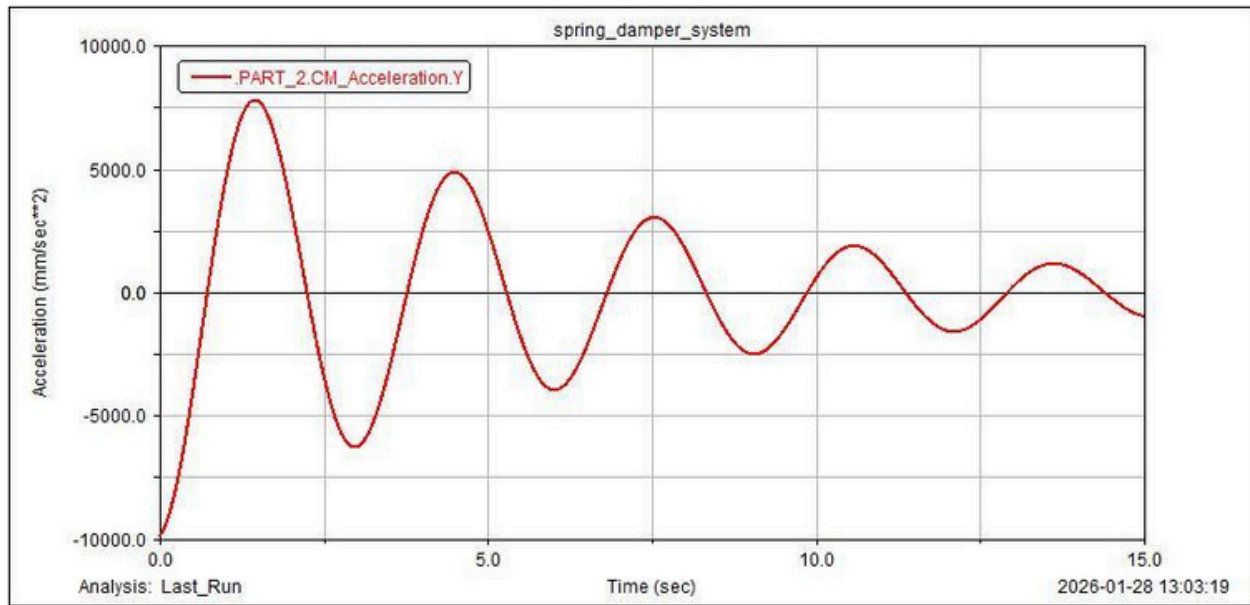
- Plot: $x'(t)$ vs Time
- Velocity reduces to zero as system stabilizes
- Without Damping Coefficient More Stiffness Coefficient



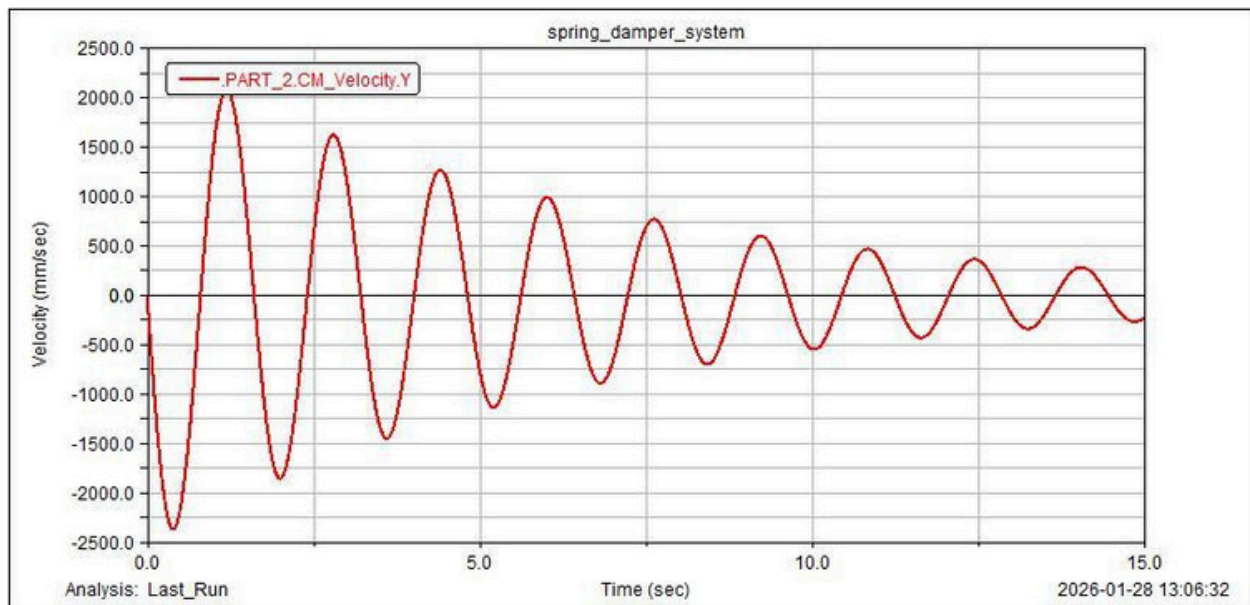
- Without Damping Coefficient Less Stiffness Coefficient



- With Damping Coefficient More Stiffness Coefficient

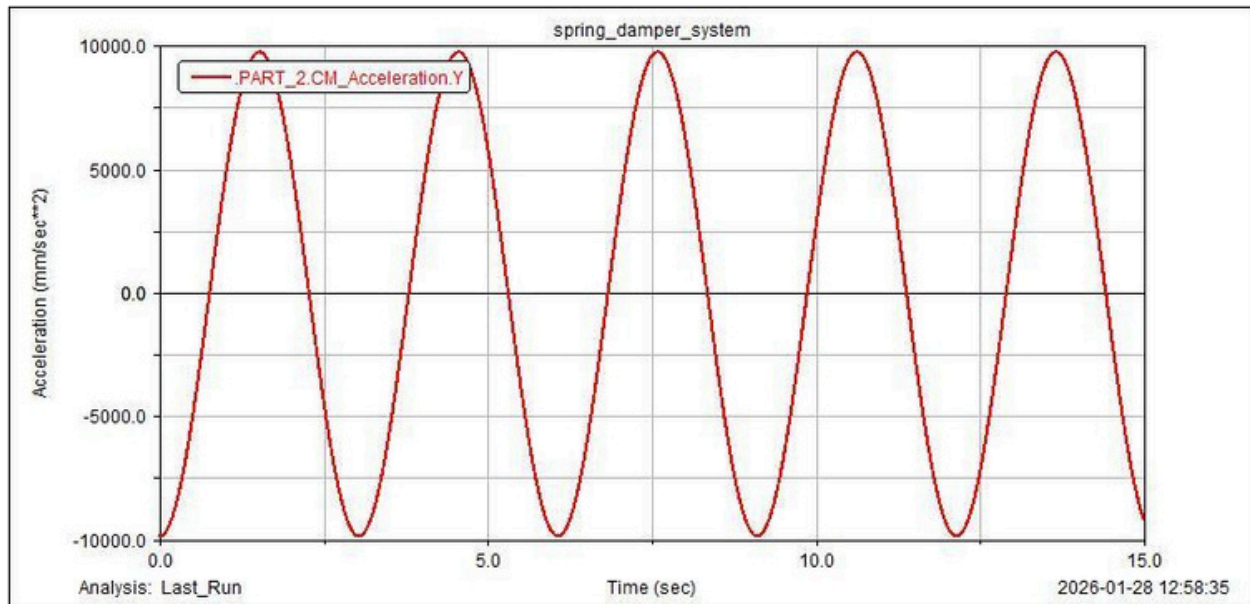


- With Damping Coefficient Less Stiffness Coefficient

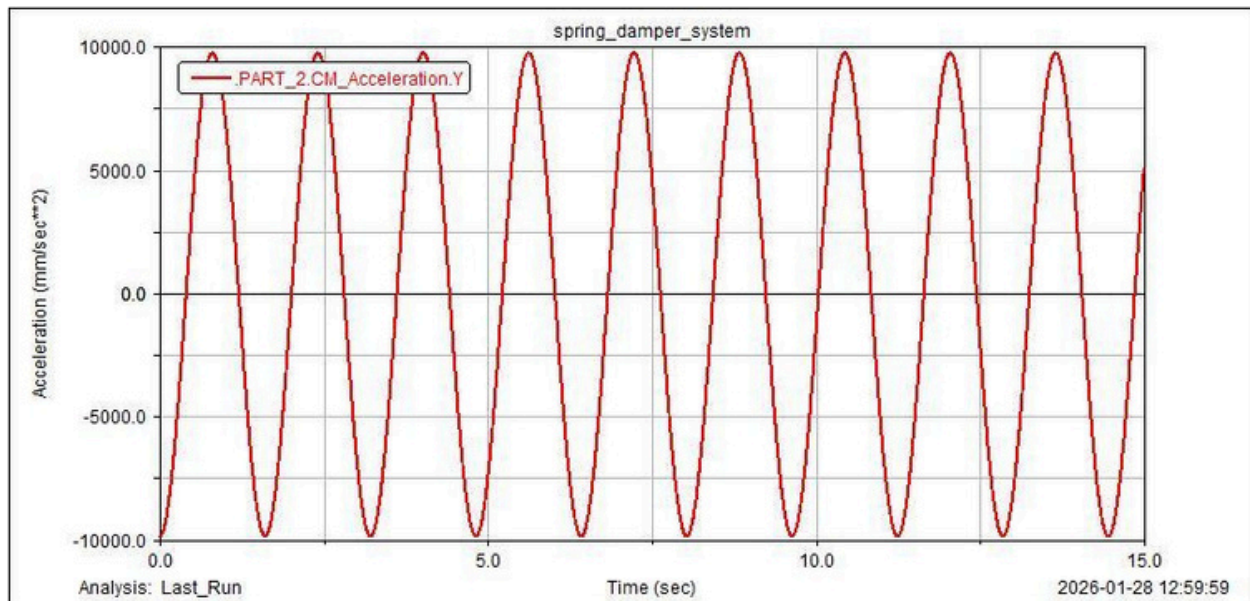


Section 3.1.3 Acceleration profile

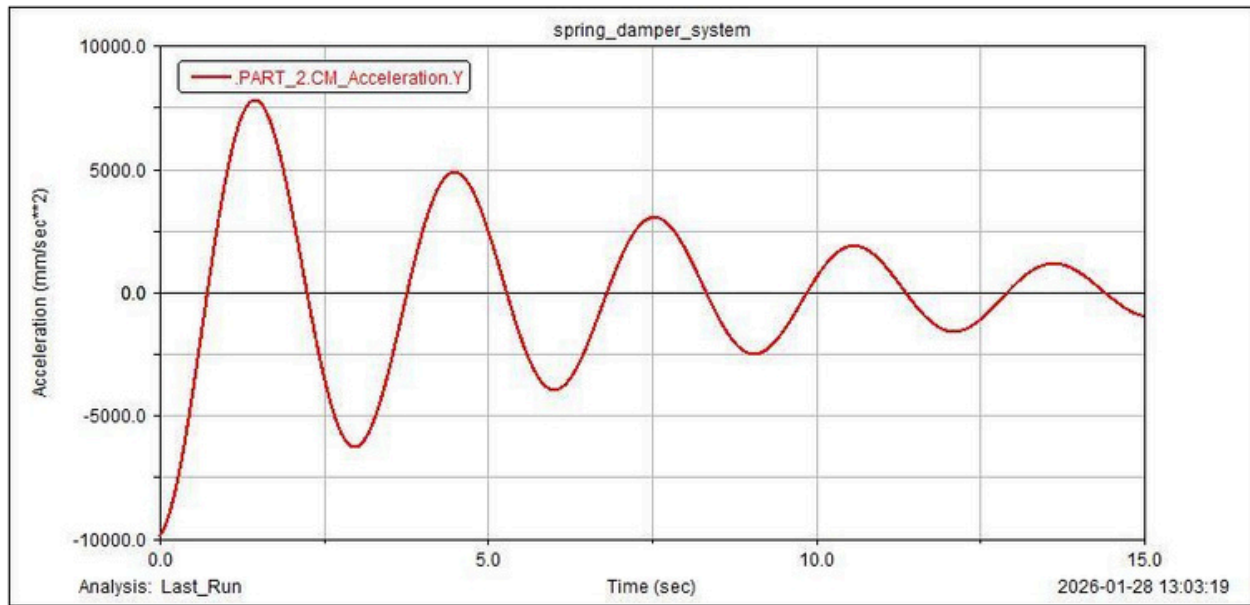
- Plot: $\ddot{x}(t)$ vs Time
- High initial acceleration, reduces with damping
- Without Damping Coefficient More Stiffness Coefficient



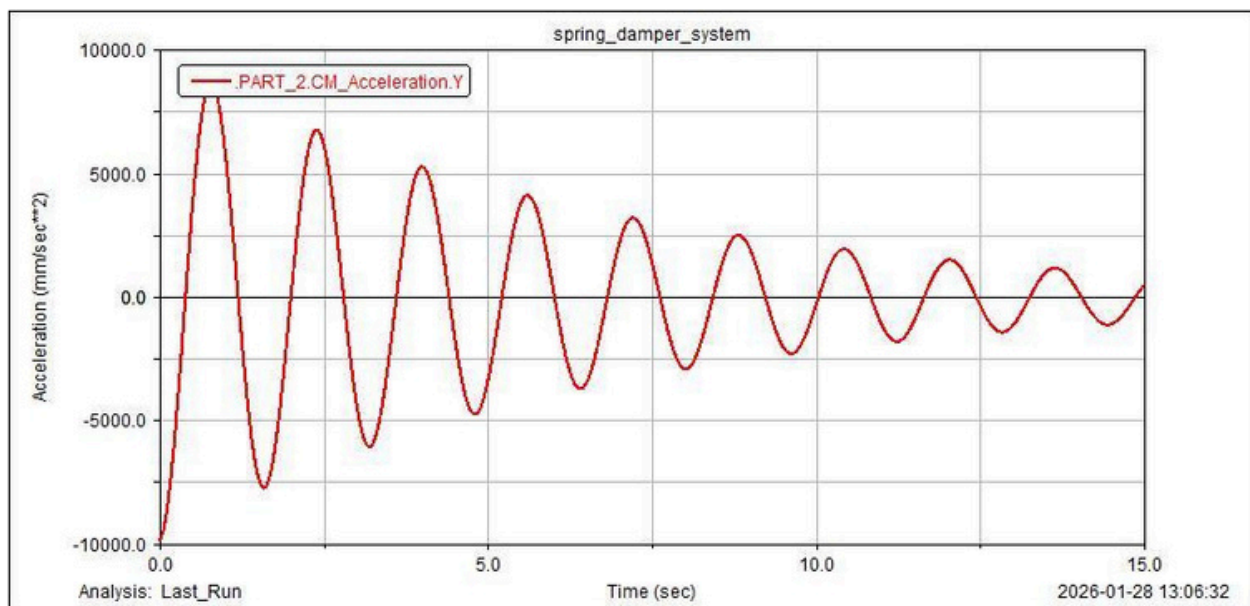
- Without Damping Coefficient Less Stiffness Coefficient



- With Damping Coefficient More Stiffness Coefficient



- With Damping Coefficient Less Stiffness Coefficient



Section 3.1.4 Comments

- Higher damping reduces oscillations faster
- Underdamped system shows oscillatory decay
- Critically damped system reaches equilibrium fastest

Section 4: Dynamic Motion Analysis

Section 4.1 Constant Driving Force / Torque

Section 4.1.1 Constant force

Let:

$$F=10 \text{ N}$$

Plot:

- Velocity vs Time
- Acceleration vs Time
(for mass, spring force, damper force)

Section 4.1.2 Double constant force

$$F=20 \text{ N}$$

- Repeat plots
- Superimpose results with Section 4.1.1
- Observe increase in peak velocity and acceleration

Section 4.2 Sinusoidal Driving Force

Section 4.2.1 Sinusoidal force

$$F(t)=1.4F_0 \sin(\omega t)$$

Minimum value = 0

Plot:

- Velocity vs Time
- Acceleration vs Time

Section 4.2.2 Double sinusoidal force

$$F(t) = 2 \times 1.4 F_0 \sin(\omega t)$$

- Superimpose plots
- Observe resonance and amplitude increase

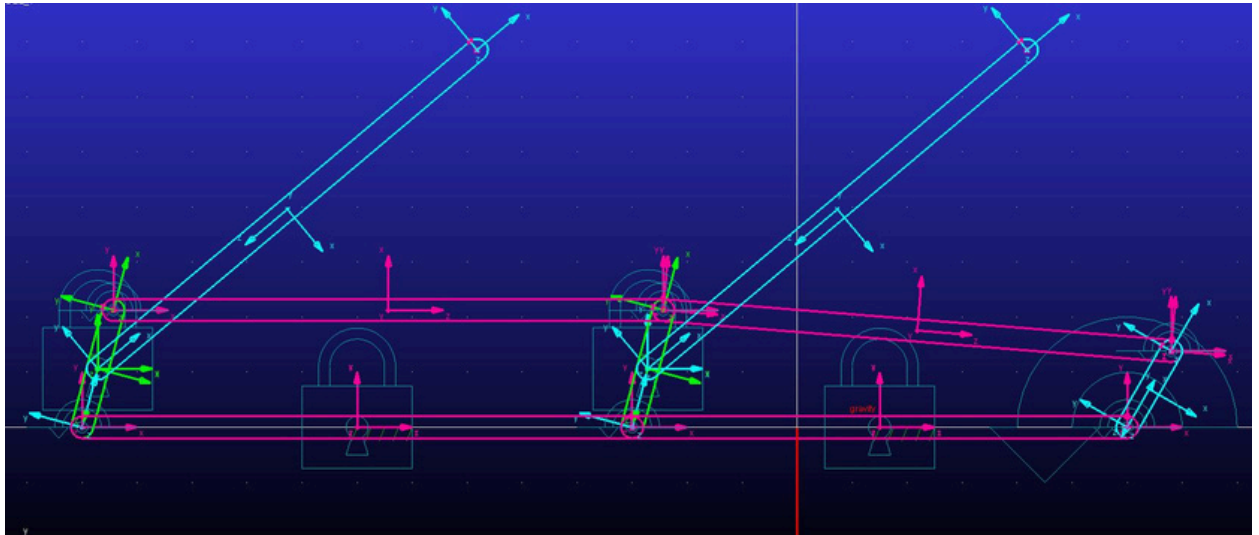
TOPIC:WIPER MECHANISM

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

The mechanism consists of:

- Electricmotor(driver)
- Crank(rotatinglink)
- Connectinglink(coupler)
- Rockerarm(wiperarm)
- Wiperbladeattachedtorocker
- Fixedframe(vehiclebody)



Section 1.2 Purpose of the mechanism

The purpose is to:

- Convert continuous rotary motion of the motor into oscillatory (reciprocating angular) motion of the wiper arm
- Sweep water/debris off the windshield over a fixed angular range

Applications:

- Automobile windshield wipers
- Rear-window wipers
- Industrial surface cleaners

Section 1.3 Kinematic sketch

- A crank-rocker four-bar linkage:
 - Ground (frame)
 - Crank (input link)
 - Coupler (connecting rod)
 - Rocker (output link)
- The rocker oscillates between two angular limits (wipe angles)

Degrees of Freedom (DOF): 1

Section 1.4 Expected operation

- Motor rotates the crank at (nearly) constant angular speed
- The coupler transmits motion to the rocker
- The rocker oscillates, producing a back-and-forth wiping motion
- The motion is non-uniform: slower near reversal, faster mid-sweep

Operating speed:

- Typically 30–60 rpm at the crank (varies by setting)

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
------------------	------------------

Crank length (r)	30mm
-------------------------	-------------

Coupler length (l)	120 mm
---------------------------	---------------

Rocker length (R)	80mm
--------------------------	-------------

Ground link distance	100 mm
-----------------------------	---------------

Wiper blade length	300 mm
---------------------------	---------------

Section 2.1.2 Masses of every component

Component	Mass
------------------	-------------

Crank	0.3 kg
--------------	---------------

Coupler	0.6 kg
----------------	---------------

Rocker + blade	1.2 kg
-----------------------	---------------

Motor rotor (equiv.)	0.5 kg
-----------------------------	---------------

Section 2.1.3 Moments of inertia

Component	Moment of Inertia
------------------	--------------------------

Crank $I = \frac{1}{2}(mr^2)$

Coupler $I = \frac{1}{12}(ml^2)$

Rocker $I \approx \frac{1}{3}(mR^2)$
(about pivot)

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- All joints are pin (revolute) joints
- Motion is planar
- Ground link is fixed
- Rocker limited to an angle sweep (e.g. $60^\circ - 120^\circ$)

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- Motor provides constant angular velocity ω or torque to the crank

Section 2.3.2 Output component

- Rocker (wiper arm) angular displacement $\theta_o(t)$
- Tip trajectory of wiper blade across windshield

Section 3: Kinematic simulation

For a four-bar, positions are obtained via loop-closure:

$$r_{\text{crank}} + r_{\text{coupler}} = r_{\text{ground}} + r_{\text{rocker}}$$

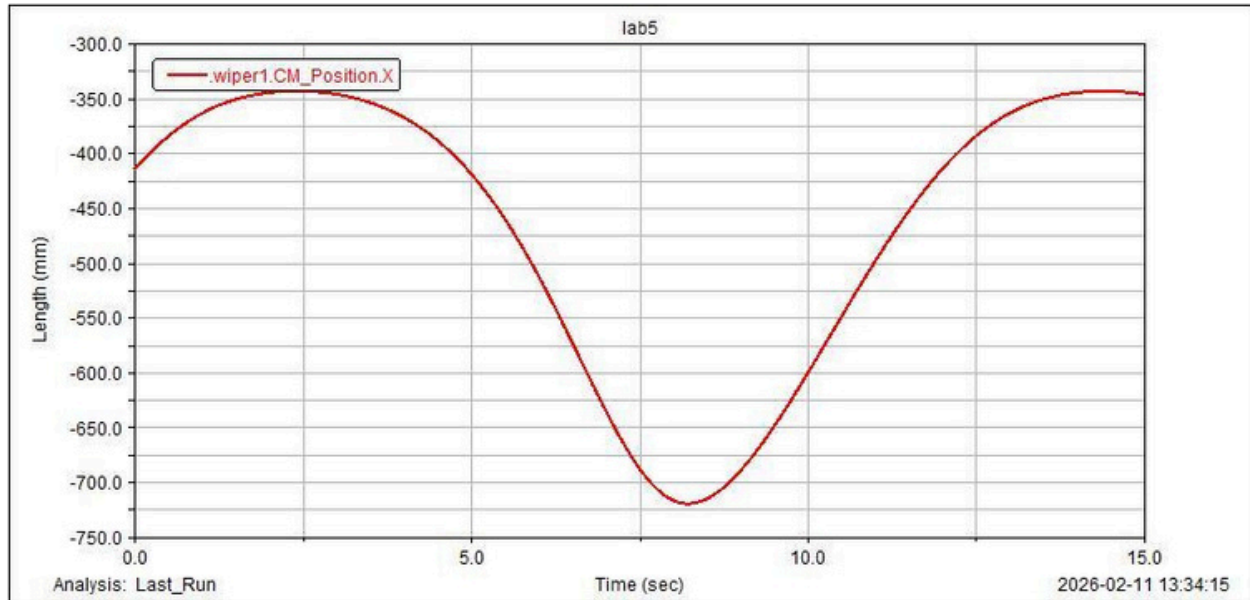
Solve for the rocker angle θ_o given crank angle θ_i .

Section 3.1 Output Component

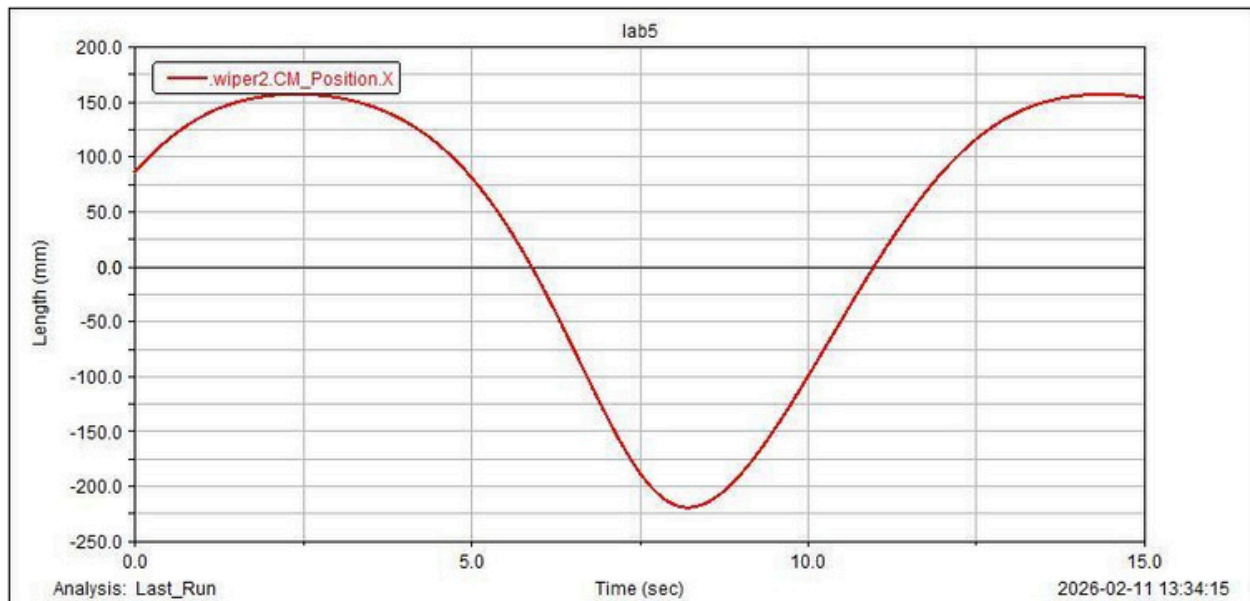
Section 3.1.1 Displacement profile

- Plot:Rockerangle θ_0 vsTime/Crankangle
- Showslimitedoscillationrange(wipeangle)

Wiper1



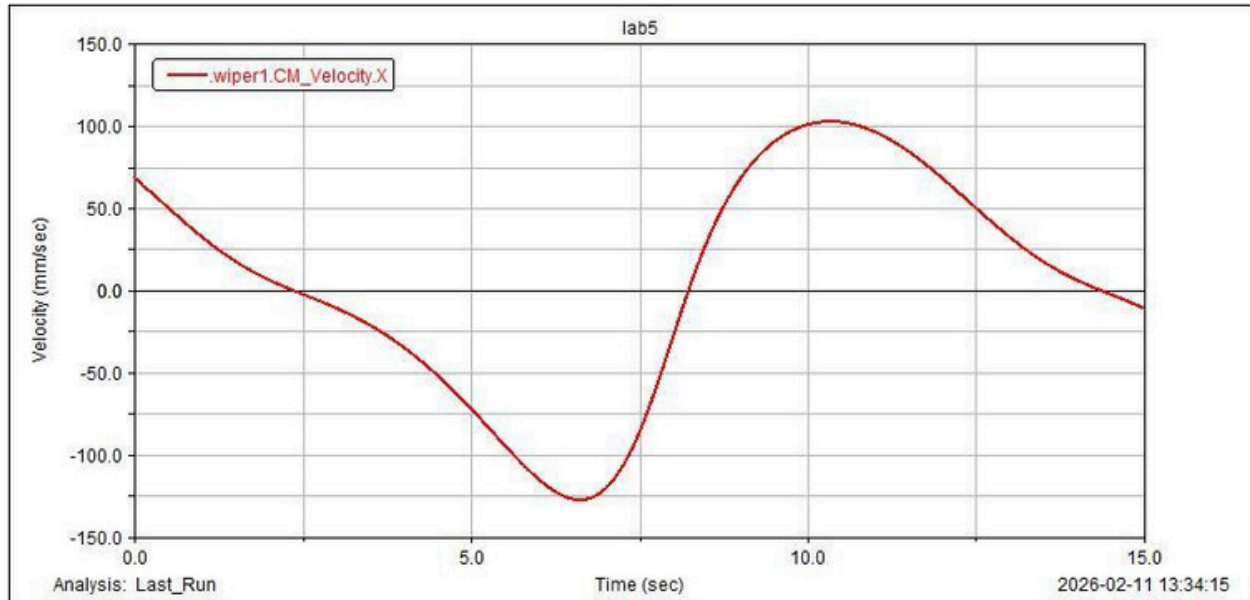
Wiper2



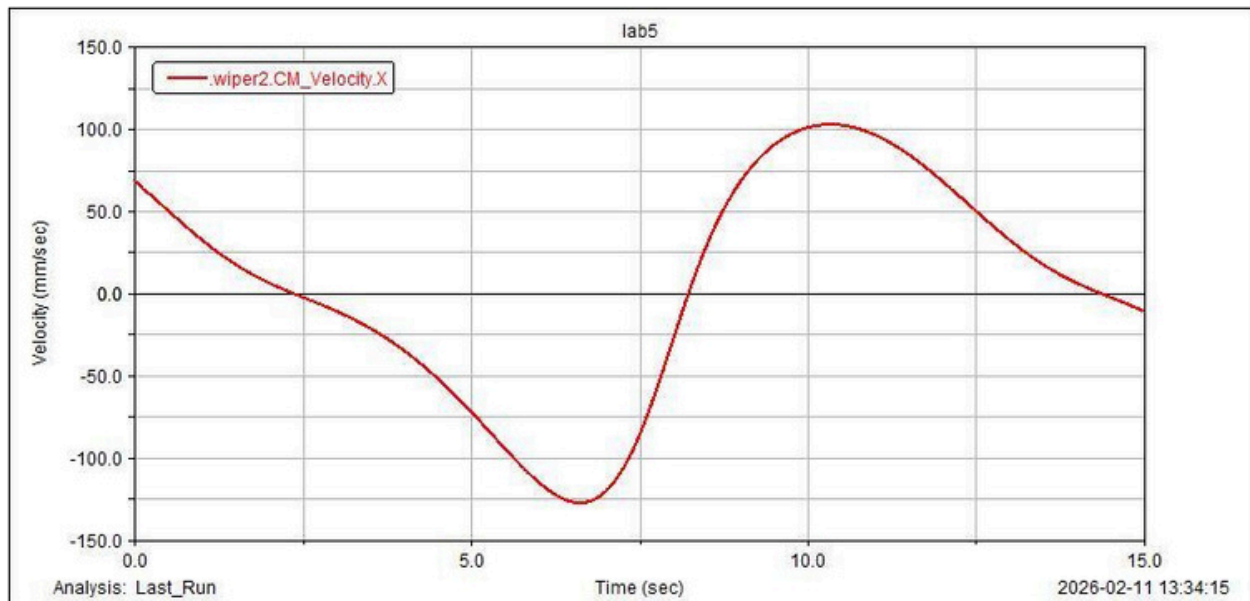
Section 3.1.2 Velocity profile

- Plot: $\dot{\theta}$ vs Time
- Non-uniform; peak velocity near mid-stroke

Wiper1



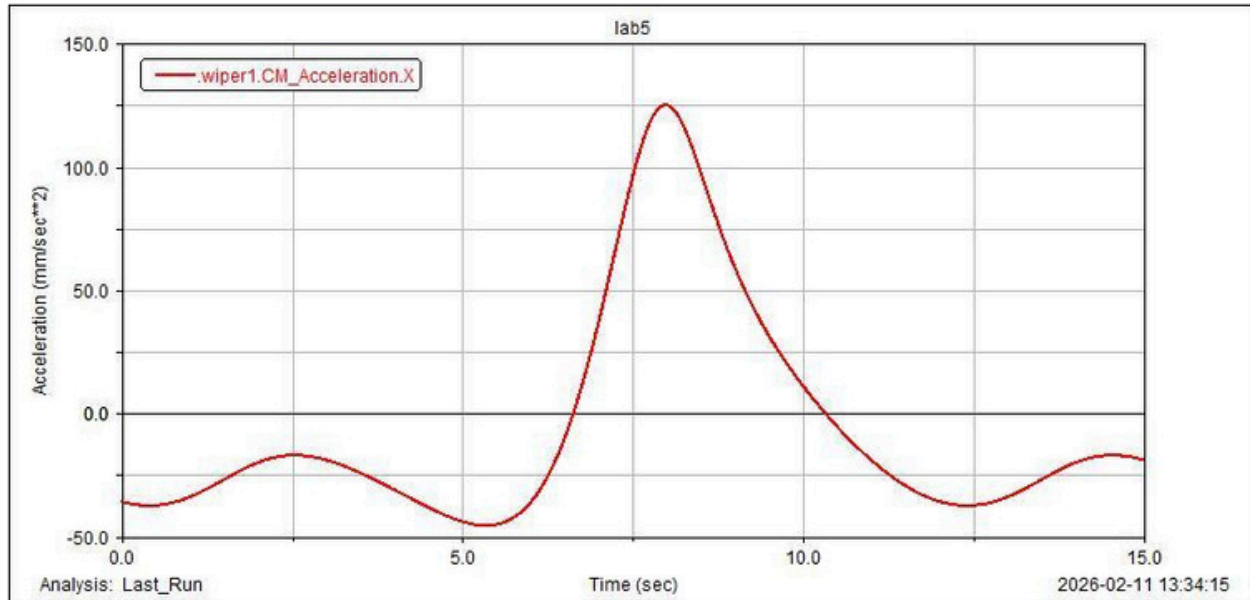
Wiper2



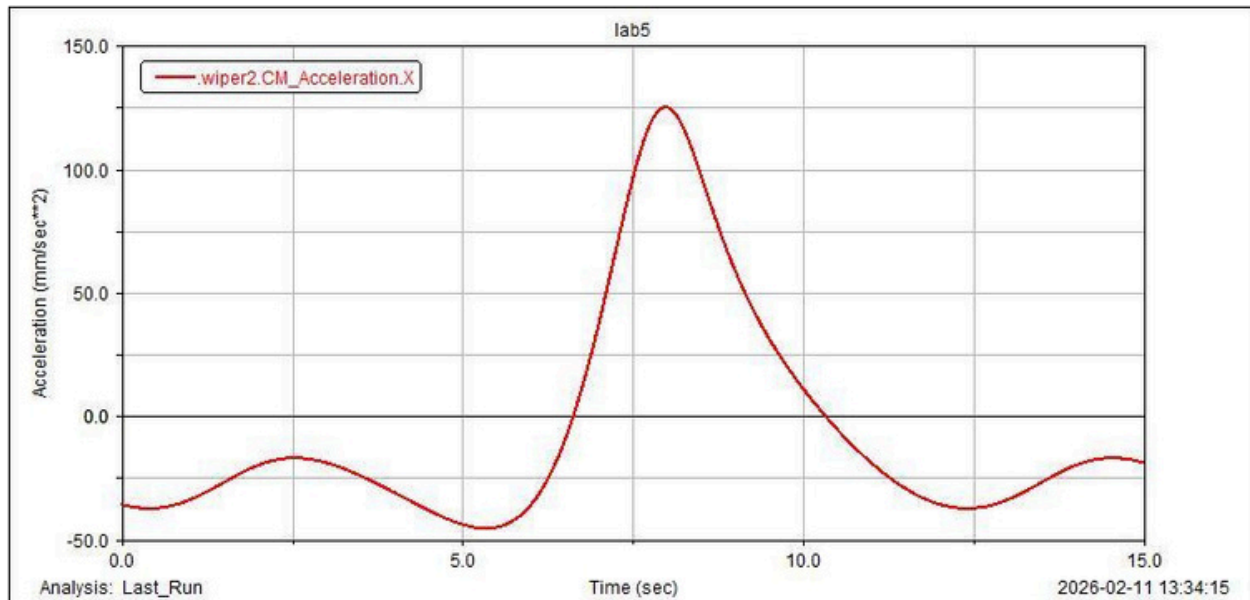
Section 3.1.3 Acceleration profile

- Plot: $\ddot{\theta}$ vs Time
- High magnitudes near reversal (end of sweep)

Wiper1



Wiper2



Section 3.1.4 Comments

- Link proportions determine wipe angle and speed profile
- Longer coupler → smoother motion
- Quick-return characteristics may appear depending on geometry

Section 4: Dynamic Motion Analysis

Section 4.1 Constant Driving Torque

Section 4.1.1 Constant torque

Let:

$$T=8 \text{ Nm}$$

Plot:

- Rocker angular velocity vs Time
- Rocker angular acceleration vs Time
- Joint reaction forces

Section 4.1.2 Double torque

$$T=16 \text{ Nm}$$

- Superimpose plots
- Observe increased peak velocities and joint loads

Section 4.2 Resistive Load (Windshield Friction / Water Drag)

Section 4.2.1 Constant resistive torque

$$T_r = 2 \text{ Nm}$$

- Apply opposing torque at rocker
- Plot velocity and acceleration vs Time

Section 4.2.2 Increased resistive torque

$$T_r = 4 \text{ Nm}$$

- Superimpose with previous case
- Observe reduced speed and possible stall at low input torque

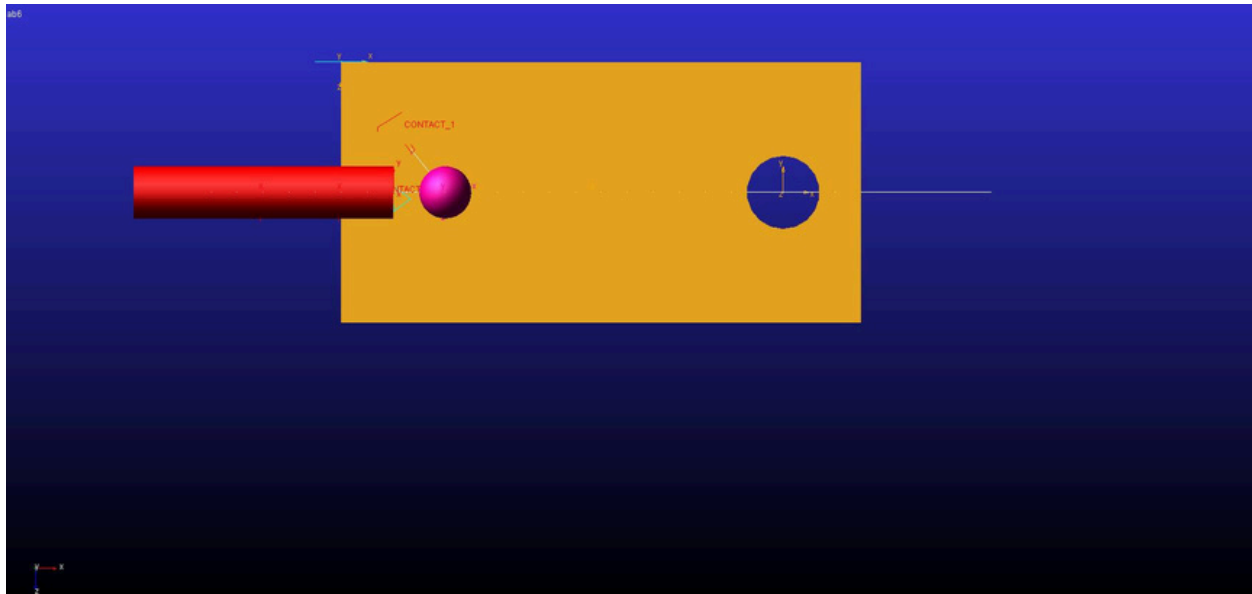
TOPIC: SNOOKER MECHANISM

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

The mechanism consists of:

- Cuestick(input actuator)
- Cueball(striker)
- Targetballs
- Flat table surface(with low friction)
- Boundaries(cushions)



Section 1.2 Purpose of the mechanism

The purpose is to:

- Demonstrate collision dynamics
- Illustrate conservation of momentum and energy
- Show force transfer between rigid bodies

Applications:

- Physics simulation of impacts
- Sports mechanics analysis

- Robotics and contact dynamics studies

Section 1.3 Kinematic sketch

- Ball treated as rigid bodies undergoing planar motion
- Motion occurs in 2D plane
- Collisions are assumed elastic or partially elastic

Degrees of Freedom (DOF):

- Each ball has 2 translational DOF (x,y)
- Total DOF = $2n$ (for n balls)

Section 1.4 Expected operation

- Cue applies impulse to cue ball
- Cue ball travels and collides with target ball(s)
- Momentum and energy are transferred
- Balls move in different directions after impact

Operating speed:

- Moderate speed impact motion

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
Ball diameter	52mm
Table size	3.6 m × 1.8 m
Cushion height	40mm

Section 2.1.2 Masses of every component

Component	Mass
------------------	-------------

Each ball	0.16 kg
------------------	----------------

Cue	~0.5 kg
------------	----------------

Table	Fixed
--------------	--------------

Section 2.1.3 Moments of inertia

Component	Moment of Inertia
------------------	--------------------------

Ball (solid sphere)	$I = \frac{2}{5}mr^2$
----------------------------	---

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- **Motion restricted to 2D plane**
- **Rolling and sliding possible**
- **Collisions along line of centers**
- **Cushion reflections follow angle laws**

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- **Cue stick applies impulse force to cue ball**

Section 2.3.2 Output component

- **Velocities of balls after collision**
- **Direction of motion**

Section 3: Kinematic simulation

Velocity after collision (1D simplified case)

$$v1 = (m1 + m2) / (m1 - m2) \quad u1$$

$$v2 = 2m1 / (m1 + m2) \quad u1$$

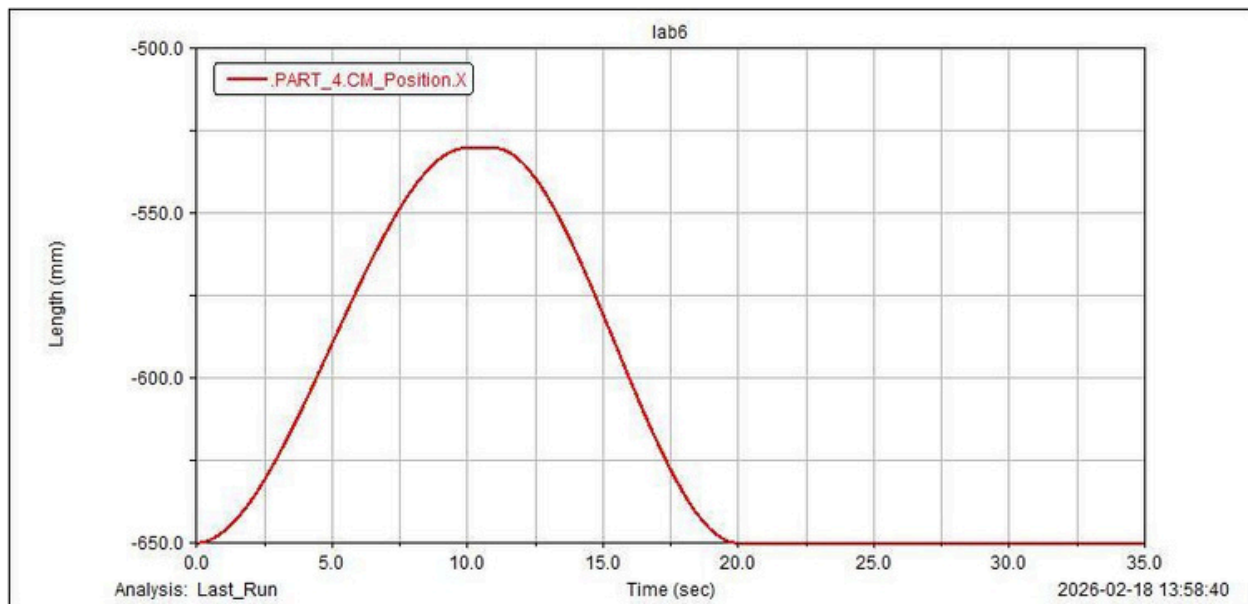
For equal masses:

$$v1=0, v2=u1$$

Section 3.1 Output Component

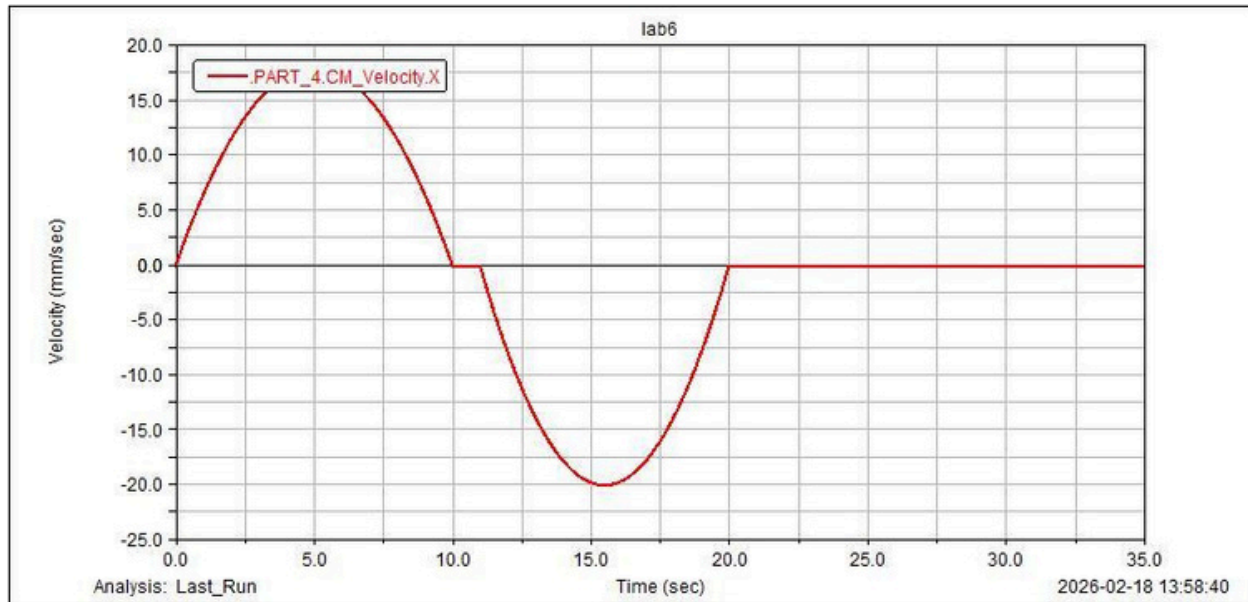
Section 3.1.1 Displacement profile

- Plot: Ball position vs Time
- Shows trajectories after collision



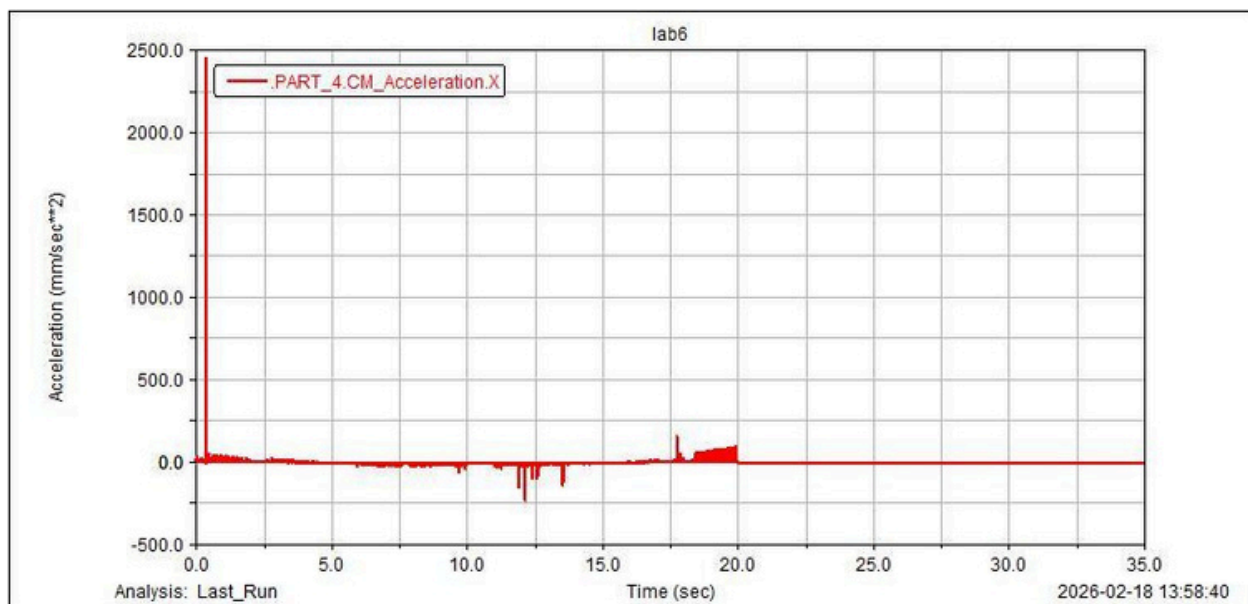
Section 3.1.2 Velocity profile

- Plot: Velocity vs Time
- Sudden changes at impact



Section 3.1.3 Acceleration profile

- Plot: Acceleration vs Time
- Impulse causes sharp peaks



Section 3.1.4 Comments

- Equal mass $\rightarrow$ complete velocity transfer
- Oblique collision $\rightarrow$ angled trajectories
- Friction reduces speed over time

Section 4: Dynamic Motion Analysis

Section 4.1 Single Collision

Section 4.1.1 One ball impact

Let: $u_1 = 2\text{m/s}$, $u_2 = 0$ Plot:

- Velocity vs Time
- Momentum vs Time

Section 4.1.2 Higher velocity impact

$u_1 = 4\text{m/s}$

- Superimpose results
- Observe higher impulse force

Section 4.2 Oblique Collision

Section 4.2.1 Angled collision

- Cue ball strikes at angle θ
- Resolve velocity into components

Section 4.2.2 Double-angle case

- Compare trajectories for different angles
- Superimpose motion paths

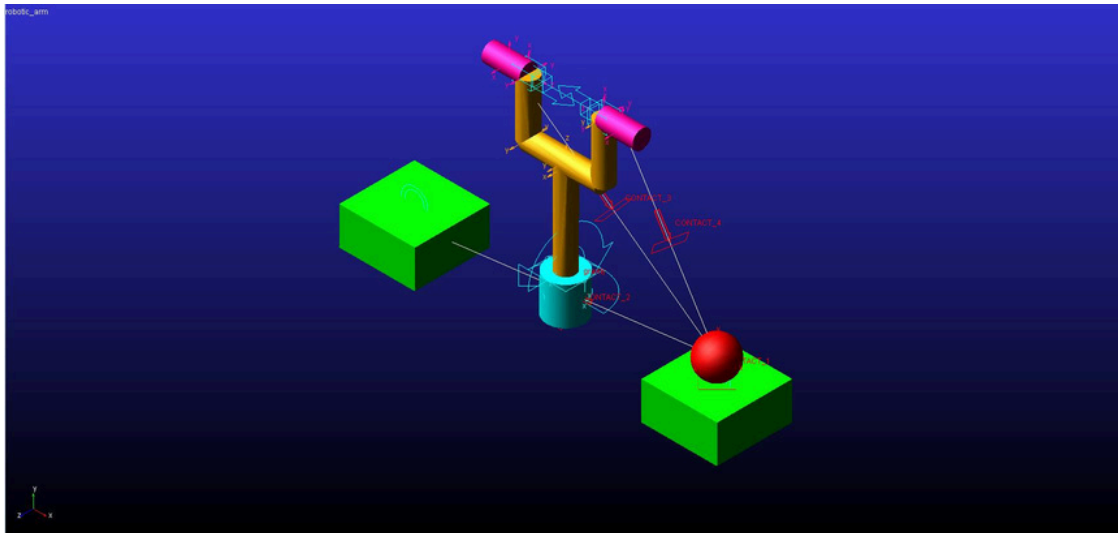
TOPIC:ROBOTIC ARM

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

The mechanism consists of:

- **Base(fixedframe)**
- **Multiplelinks(rigidbodies)**
- **Joints(revolute/prismatic)**
- **Endeffector(gripper/tool)**
- **Actuators(motors/servos)**



Section 1.2 Purpose of the mechanism

The purpose is to:

- **Perform precise positioning and manipulation**
- **Convert actuator inputs into controlled motion of an end effector**
- **Execute tasks such as pick-and-place, welding, assembly**

Applications:

- **Industrial automation**
- **Medical robotics**
- **Material handling**

- Space and service robots

Section 1.3 Kinematic sketch

- A serial chain of links connected by joints (typically RRR or RRP)
- Motion described using joint variables (angles/displacements)
- End effector position obtained via forward kinematics

Degrees of Freedom (DOF):

- Equal to number of independent joints (e.g., 3DOF for a 3-link planar arm)

Section 1.4 Expected operation

- Actuators drive joint motions
- Link motions combine to position the end effector in the workspace
- Motion can be point-to-point or continuous path tracking

Operating speed:

- Moderate to high, depending on application and payload

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
Link 1 length (L1)	200 mm
Link 2 length (L2)	150 mm
Link 3 length (L3)	100 mm

Base height 100 mm

Section 2.1.2 Masses of every component

Component	Mass
------------------	-------------

Link 1	2kg
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Link 2	1.5 kg
---------------	---------------

Link 3	1kg
---------------	------------

End effector	0.5 kg
-------------------------	---------------

Section 2.1.3 Moments of inertia

Component	Moment of Inertia
------------------	--------------------------

Link 1	$I=1/12mL^2$
---------------	--------------------------------

Link 2	$I=1/12mL^2$
---------------	--------------------------------

Link 3	$I=1/12mL^2$
---------------	--------------------------------

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- **Joints allow rotation (revolute joints)**
- **Motion confined to a plane (for planar arm)**
- **Base is fixed**

- Joint limits restrict motion range

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- Electric motors apply torque at each joint

Section 2.3.2 Output component

- End effector position (x,y)(x, y)(x,y)
- End effector velocity and acceleration

Section 3: Kinematic simulation

Forward kinematics (2D 3-link arm)

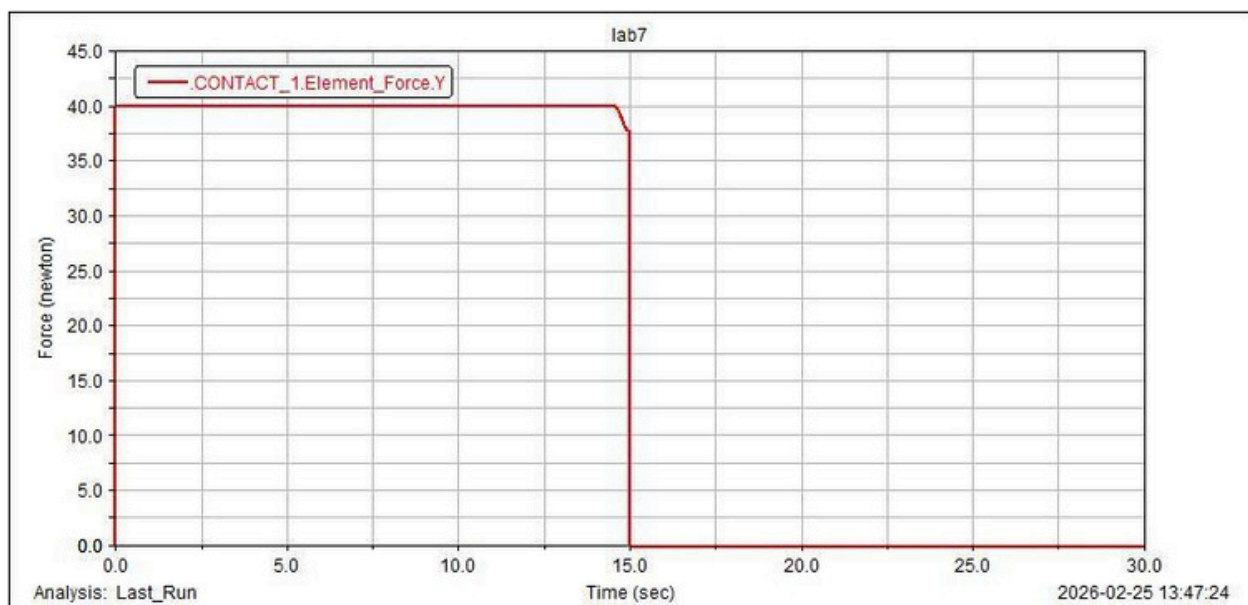
$$x=L1\cos\theta1+L2\cos(\theta1+\theta2)+L3\cos(\theta1+\theta2+\theta3)$$

$$y=L1\sin\theta1+L2\sin(\theta1+\theta2)+L3\sin(\theta1+\theta2+\theta3)$$

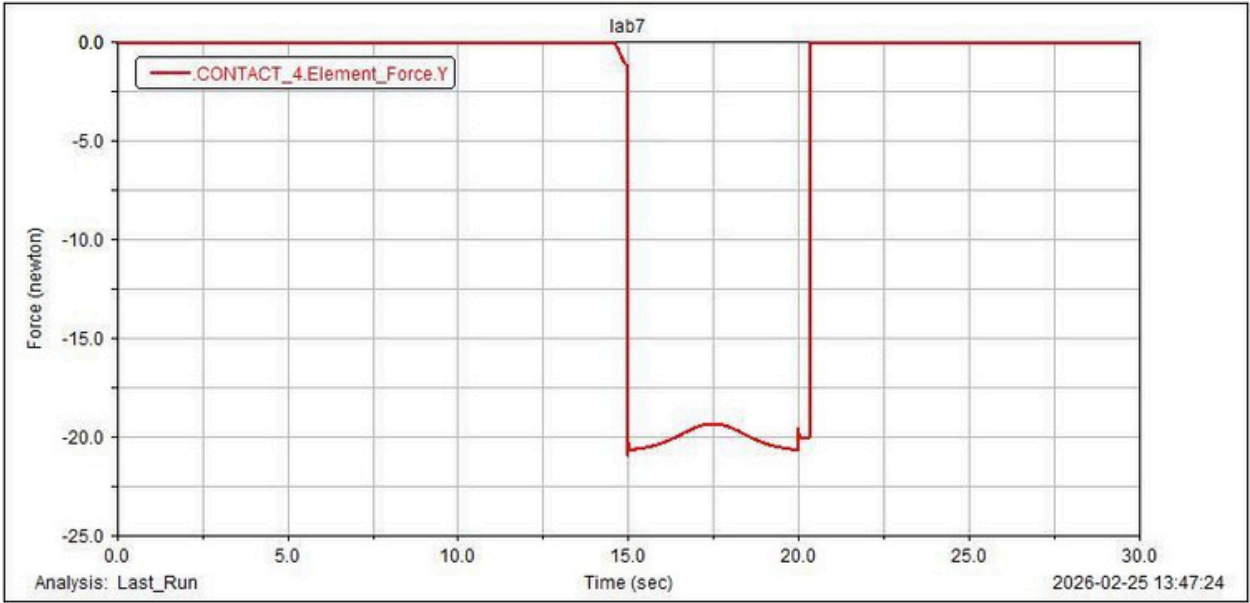
Section 3.1 Output Component

Section 3.1.1 Contact Force profile

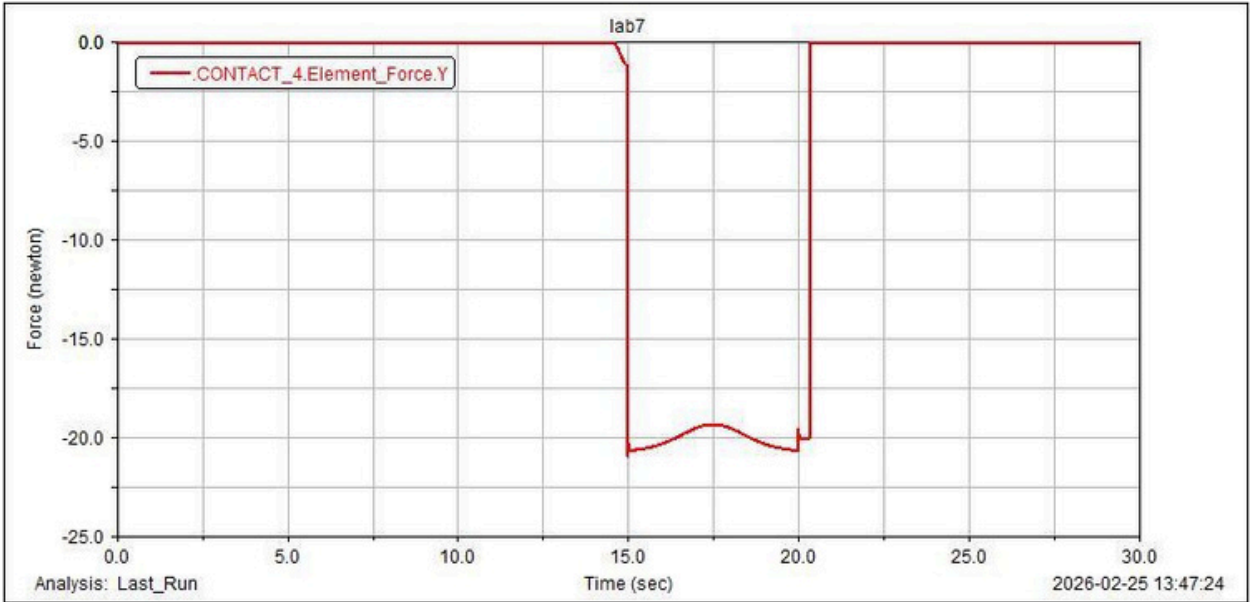
Contact Force 1



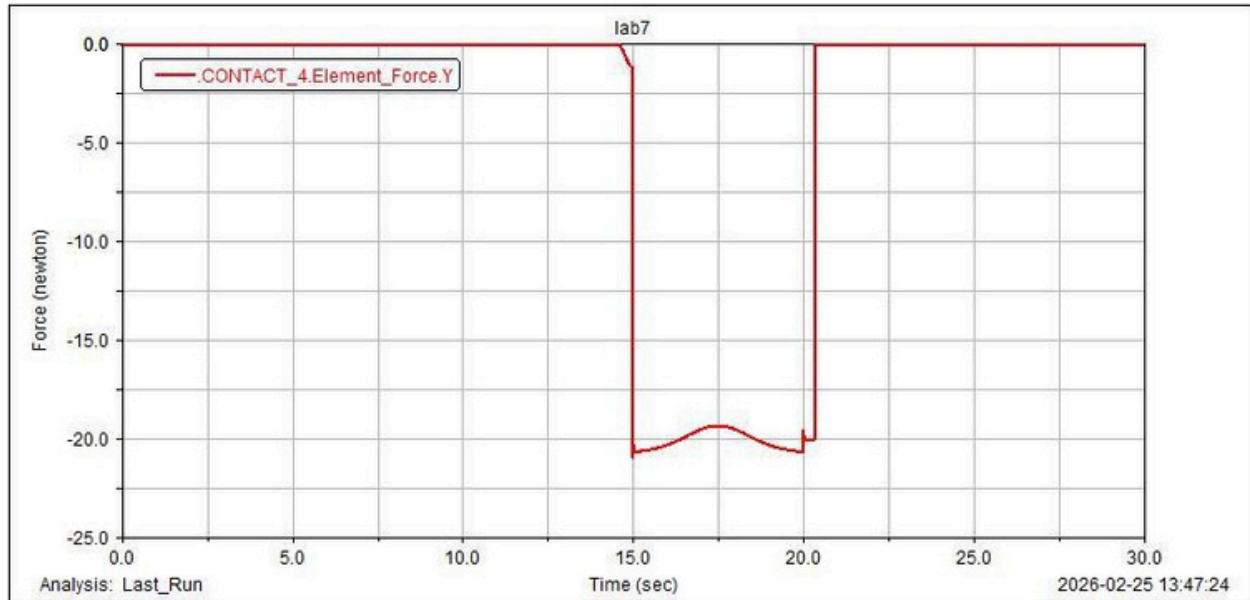
Contact Force 2



Contact Force 3



Contact Force 4



Section 3.1.2 Comments

- Nonlinear relationship between joint space and Cartesian space
- Singularities may occur at certain configurations
- Workspace limited by link lengths

Section 4: Dynamic Motion Analysis

Section 4.1 Constant Joint Torque

Section 4.1.1 Constant torque

Let:

$$T_1 = T_2 = T_3 = 5 \text{ Nm}$$

Plot:

- Joint velocities vs Time
- End effector velocity vs Time

Section 4.1.2 Double torque

$$T = 10 \text{ Nm}$$

- Superimpose plots
- Observe increased speed and acceleration

Section 4.2 Trajectory-Based Motion

Section 4.2.1 Sinusoidal joint motion

$$\theta_i(t) = \theta_0 \sin(\omega t)$$

Plot:

- End effector path
- Velocity and acceleration profiles

Section 4.2.2 Increased amplitude

$$\theta_i(t) = 2\theta_0 \sin(\omega t)$$

Superimpose trajectories

- Observe expanded workspace coverage

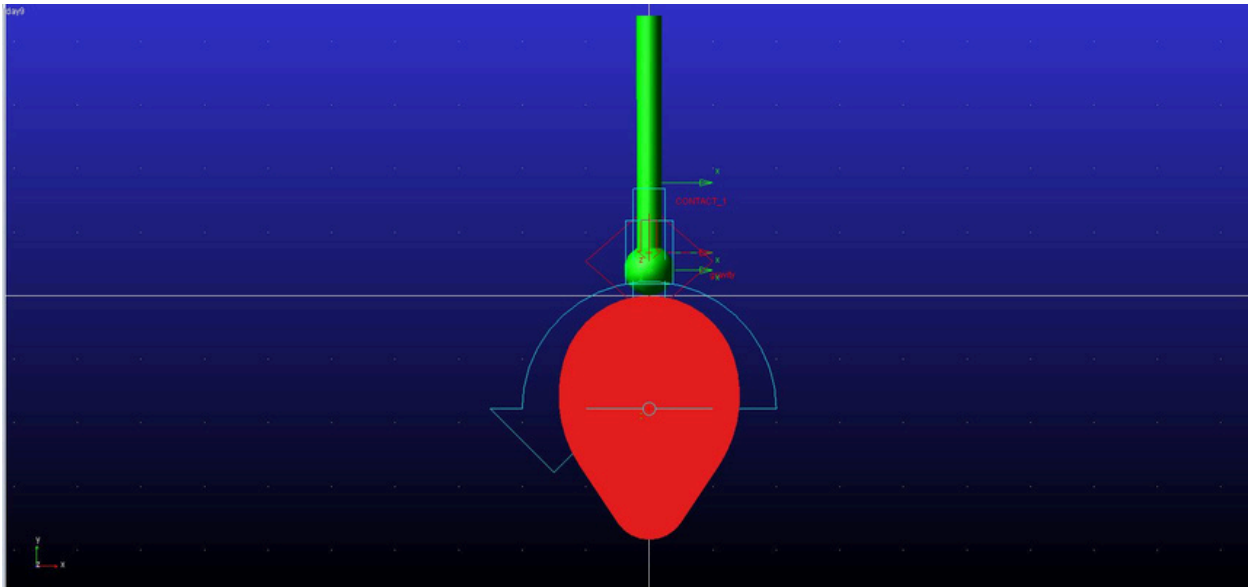
TOPIC:CAM AND FOLLOWER MECHANISM

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

The mechanism consists of:

- Cam(rotatingdriverwithadesignedprofile)
- Follower(translatingoroscillatingelement)
- Guideforfollowermotion
- Springorgravityformaintainingcontact



Section 1.2 Purpose of the mechanism

The purpose is to:

- Convert rotary motion into a prescribed follower motion
- Achieves specific displacement, velocity, and acceleration profiles

Applications:

- Internal combustion engines (valve actuation)
- Textile machines
- Packaging and automation systems

Section 1.3 Kinematic sketch

- Cam rotates about a fixed axis
- Follower moves in a constrained path (linear or oscillating)
- Motion defined by cam profile geometry

Degrees of Freedom (DOF): 1

Section 1.4 Expected operation

- Cam rotates continuously
- Follower rises, dwells, and returns based on cam shape
- Motion phases:
 - Rise (lift)
 - Dwell
 - Return
 - Dwell

Operating speed:

- Moderate to high (depends on application)

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
-----------	-----------

Base circle radius	40mm
--------------------	------

Lift (stroke)	20mm
---------------	------

Cam thickness	10mm
---------------	------

Follower diameter	20mm
-------------------	------

Section 2.1.2 Masses of every component

Component	Mass
Cam	1.5 kg
Follower	0.5 kg
Spring	Negligible

Section 2.1.3 Moments of inertia

Component	Moment of Inertia
Cam	$I = \frac{1}{2}mr^2$
Follower	Neglected (translation dominant)

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- Cam rotates about fixed axis
- Follower constrained to linear motion
- Continuous contact maintained by spring
- No slipping between cam and follower

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- Cam driven by motor with angular velocity ω

Section 2.3.2 Output component

- Follower displacement $y(t)$
- Velocity and acceleration profiles

Section 3: Kinematic simulation

Follower displacement (example: simple harmonic motion)

$$y = h/2 (1 - \cos\theta)$$

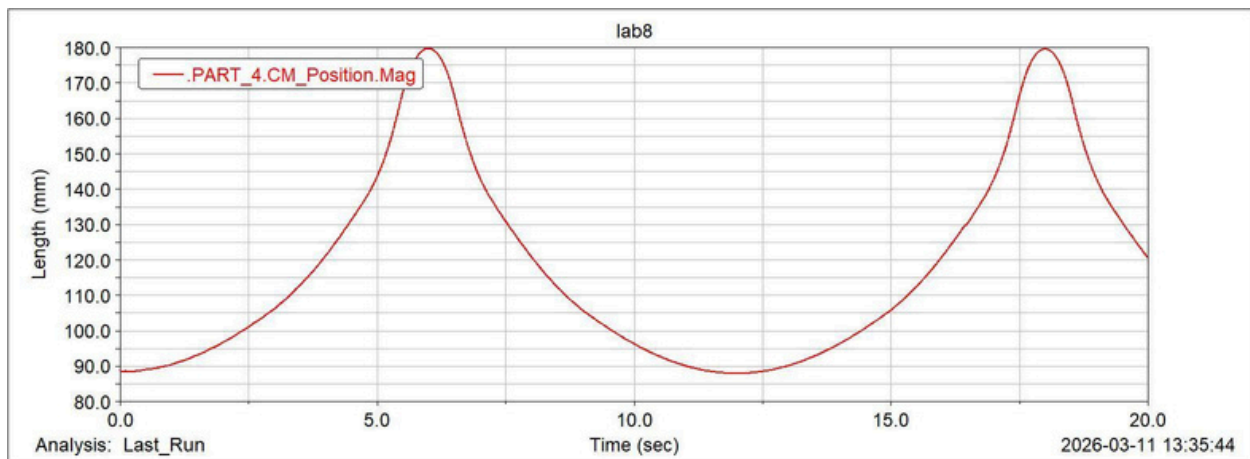
Where:

- h = lift
- θ = cam angle

Section 3.1 Output Component

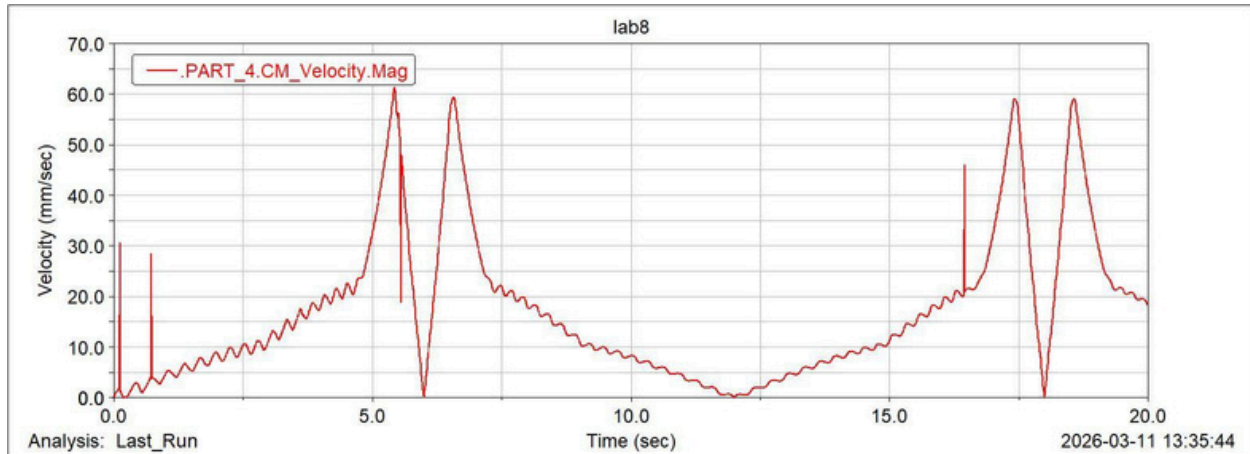
Section 3.1.1 Displacement profile

- Plot: Follower displacement vs Cam angle/time
- Shows rise, dwell, and return



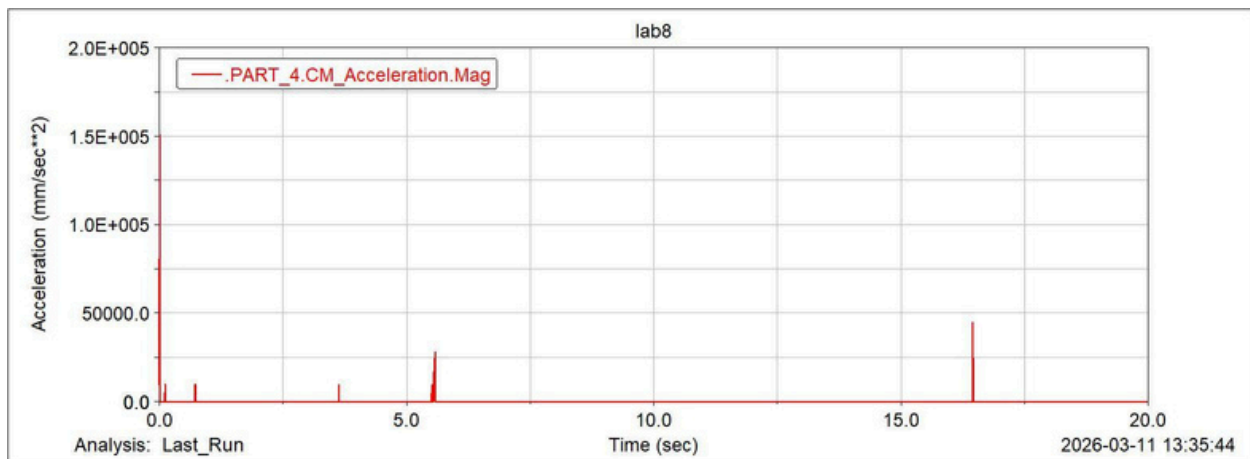
Section 3.1.2 Velocity profile

- Plot: Follower velocity vs Time
- Smooth curve for SHM profile



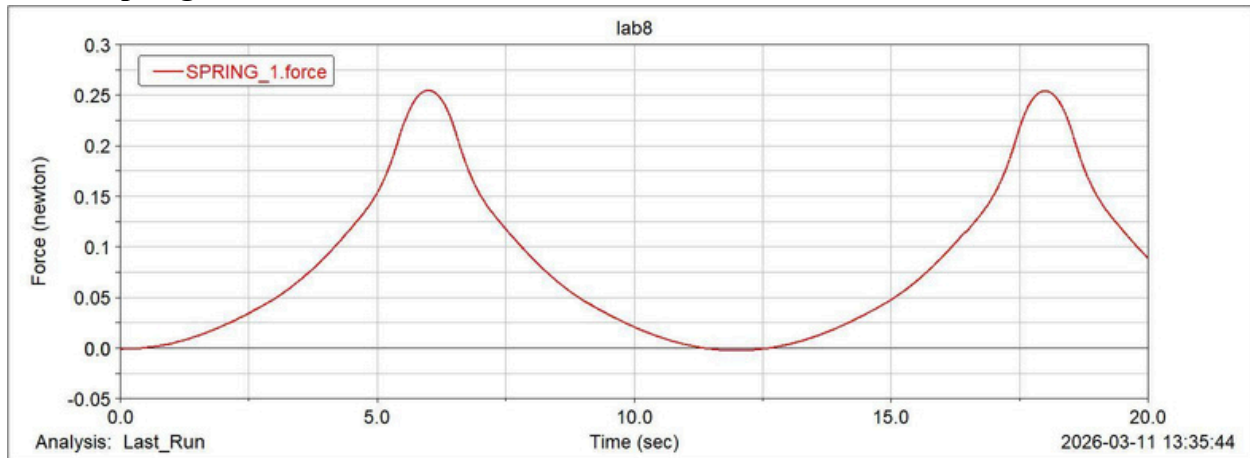
Section 3.1.3 Acceleration profile

- Plot: Follower acceleration vs Time
- Peaks during rise and return



Section 3.1.4 Force

- Spring Force



Section 3.1.5 Comments

- Cam profile determines motion characteristics
- Smooth profiles reduce vibration and wear
- High acceleration leads to higher dynamic forces

Section 4: Dynamic Motion Analysis

Section 4.1 Constant Angular Speed

Section 4.1.1 Constant speed

Let:

$$\omega = 10 \text{ rad/s}$$

Plot:

- FollowervelocityvsTime
- AccelerationvsTime
- ContactforcevsTime

Section 4.1.2 Double speed

$\omega=20$ rad/s

- Superimpose plots
- Observe increased acceleration and contact forces

Section 4.2 Different Motion Profiles

Section 4.2.1 SHM vs Uniform motion

- Compare displacement, velocity, acceleration
- SHM gives smoother motion

Section 4.2.2 Higher lift

$h=40$ mm

- Superimpose results
- Observe increased displacement and force

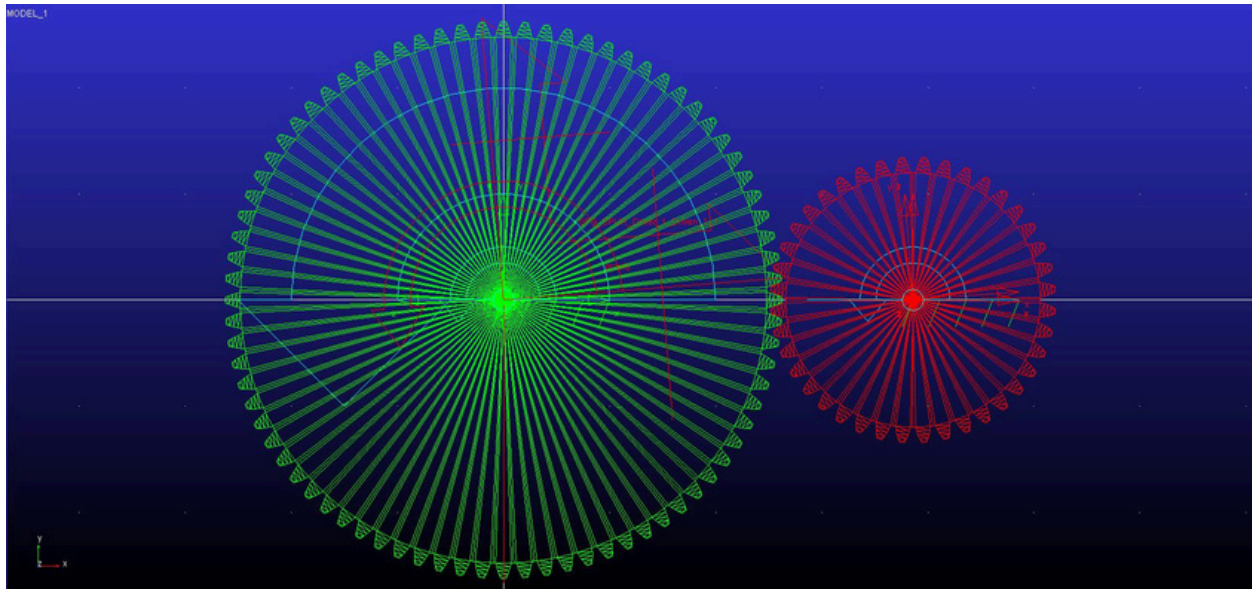
TOPIC:GEAR

Section 1: Mechanism Description

Section 1.1 Picture (or sketch) of the mechanism

The mechanism consists of:

- **Drivinggear(inputgear)**
- **Drivengear(outputgear)**
- **Gearteeth(involuteprofile)**
- **Shaftsandbearings**
- **Frame/support**



Section 1.2 Purpose of the mechanism

The purpose is to:

- Transmit motion and power between shafts
- Change speed, torque, and direction of rotation
- Provide precise velocity ratios

Applications:

- Automobiles (gearboxes)
- Industrial machinery
- Clocks and watches
- Robotic systems

Section 1.3 Kinematic sketch

- Two or more gears in mesh
- Contact between teeth ensures motion transfer
- Motion is rotational

Degrees of Freedom (DOF): 1

Section 1.4 Expected operation

- Drive gear rotates and transmits motion to driven gear
- Driven gear rotates in opposite direction (for external gears)
- Speed and torque depend on gear ratio

Operating speed:

- Low to very high (depends on design)

Section 2: Model specification

Section 2.1 Geometry and Material

Section 2.1.1 Dimensions of every component

Component	Dimension
Number of teeth (driver) Z1	20
Number of teeth (driven) Z2	40
Module (m)	2mm
Pitch diameter $D=mZ$	40 mm, 80 mm

Section 2.1.2 Masses of every component

Component	Mass
Driver gear	1kg
Driven gear	2kg
Shaft	Negligible

Section 2.1.3 Moments of inertia

Component	Moment of Inertia
Gear (solid disk approx.)	$I=1/2mr^2$

Section 2.2 Constraints

Section 2.2.1 Constraints on motion

- Gears connected via **tooth contact**

- Noslippingatpitchcircle
- Shaftsarefixedinposition
- Motionrestrictedtorotation

Section 2.3 Motion Transmission

Section 2.3.1 Driving component

- Drivegearsuppliedwithangularvelocity ω_1 ortorque

Section 2.3.2 Output component

- Drivengearangularvelocity ω_2

Section 3: Kinematic simulation

Velocity ratio

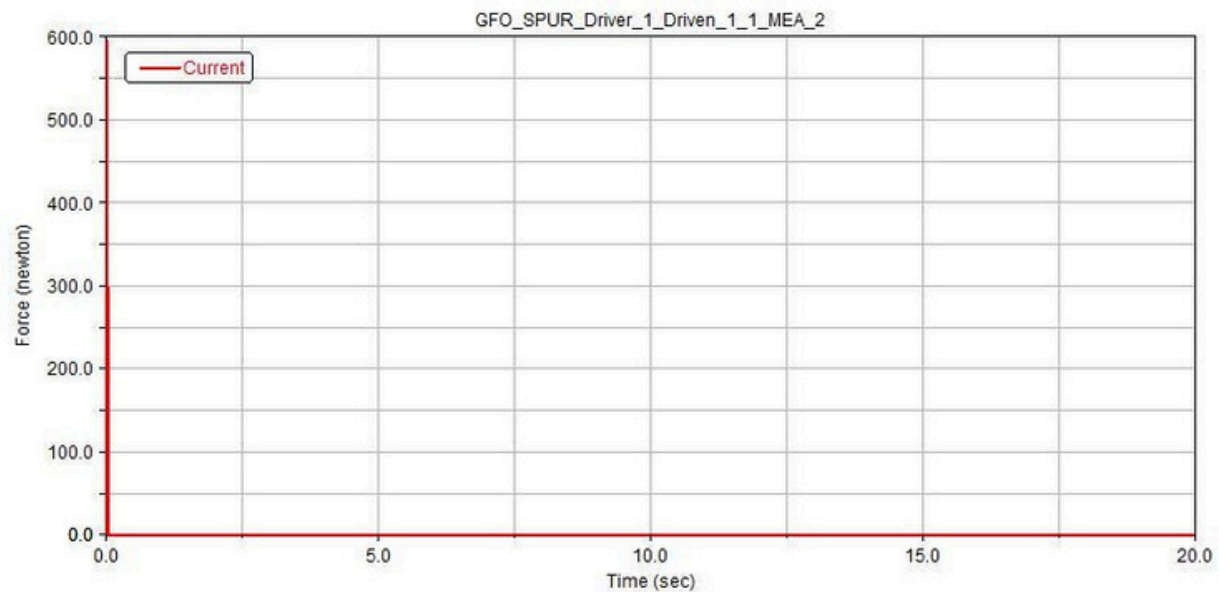
$$\omega_1/\omega_2=Z_2/Z_1$$

For given values:

$$\omega_1/\omega_2=40/20=2$$

Section 3.1 Output Component

Section 3.1.1 Force



Section 3.1.2 Comments

- Larger driven gear → lower speed, higher torque
- Motion transfer is smooth due to involute teeth
- Direction reverses for external gears

Section 4: Dynamic Motion Analysis

Section 4.1 Constant Input Torque

Section 4.1.1 Constant torque

Let:

$$T_1 = 10 \text{ Nm}$$

Output torque:

$$T_2 = Z_2/Z_1 \cdot T_1 = 20 \text{ Nm}$$

Plot:

- Angular velocity vs Time
- Torque vs Time

Section 4.1.2 Double torque

$$T_1 = 20 \text{ Nm}$$

- Superimpose plots
- Observe proportional increase in output torque

Section 4.2 Multi-stage Gear Train

Section 4.2.1 Compound gear train

- Multiple gears connected
- Overall ratio = product of individual ratios

Section 4.2.2 Increased ratio

- Compare single-stage vs multi-stage
- Observe larger speed reduction